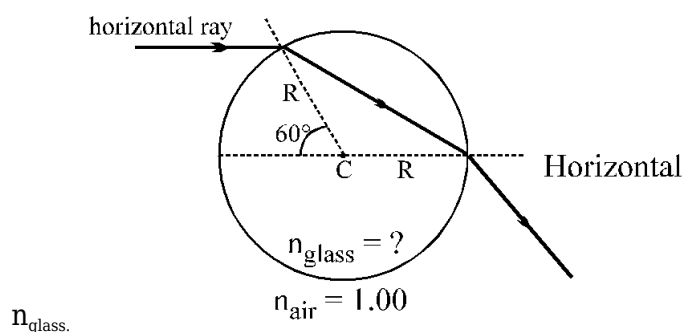


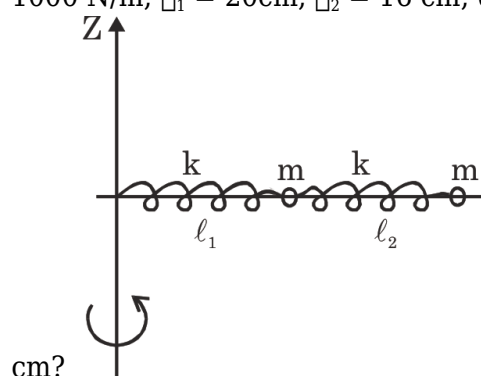
PART 1 : PHYSICS

SECTION-II

1) A ray of light, initially horizontal, passes through a cylinder of glass of radius R , as shown. Assume that $n_{\text{air}} = 1$. Based on the geometry of the ray's path as shown, find index of refraction of the glass,



2) Two identical light springs of spring constant k and two balls of same mass m are threaded on a smooth rod and rotated slowly about the vertical z axis as shown. In the steady state, the angular velocity of the system is ω and the length of the springs increase to ℓ_1 and ℓ_2 as shown. Take $k = 1000 \text{ N/m}$, $\ell_1 = 20 \text{ cm}$, $\ell_2 = 16 \text{ cm}$, $\omega = 20 \text{ rad/s}$, what is the natural length of each of the springs in

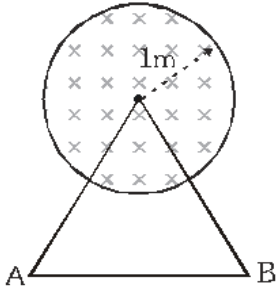


3) An engine is used to move a mass of 0.8 kg starting from rest a distance of 2 m in a straight line along a frictionless flat surface. Engine exerts a constant force of 5 N . Find the average power (in W) exerted by the engine.

4) In a Young's double slit experiment, the slits are 2 mm apart and are illuminated with a mixture of two wavelength $\lambda_0 = 750 \text{ nm}$ and $\lambda = 900 \text{ nm}$. The minimum distance (in mm) from the common central bright fringe on a screen 2 m from the slits where a bright fringe from one interference pattern coincides with a bright fringe from the other, is :-

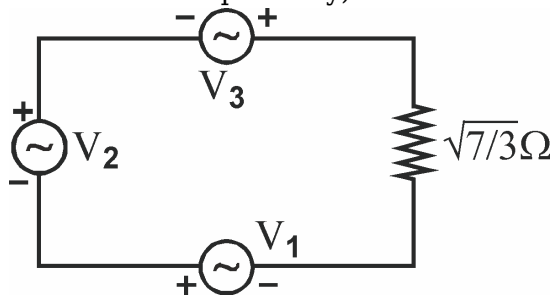
5) A Bohr hydrogen atom undergoes a transition $n = 5 \rightarrow n = 4$ and emits a photon of frequency ν . Frequency of circular motion of electron in $n = 4$ orbit is ν_4 . Find the ratio ν/ν_4 .

6) A uniform magnetic field of intensity $B = \frac{6}{\pi} \sin 2t$ directed into the plane of the paper exists in the cylindrical region of radius 1 m. A uniform loop of resistance 10Ω folded in the form of equilateral triangle of side length 2m is placed as shown in figure. Find the maximum potential drop (in volt) in wire AB.



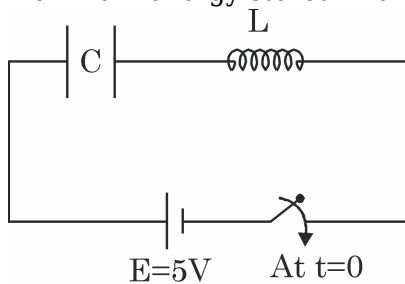
7) Three alternating voltage sources $V_1 = 3\sin\omega t$ volt, $V_2 = 5\sin(\omega t + \phi_1)$ volt and $V_3 = 5\sin(\omega t - \phi_2)$

volt are connected across a resistance $R = \sqrt{\frac{7}{3}}\Omega$ as shown in the figure (where ϕ_1 and ϕ_2 corresponds to 30° and 127° respectively). Find the rms current (in Amp) through the resistor.



[Take $\sqrt{2} = 1.4$]

8) An initially uncharged capacitor and inductor are connected to battery at $t = 0$. Find the maximum energy stored in the inductor in Joule ? [Take $C = 1\text{F}$, $L = 1\text{H}$ and $E = 5\text{V}$]



SECTION-I (i)

1)

A motor boat is to reach at an angle 120° with downstream on other side of the river velocity of boat with respect to water is $5\sqrt{3}$ m/s and velocity of river flow is 5 m/s. Then :

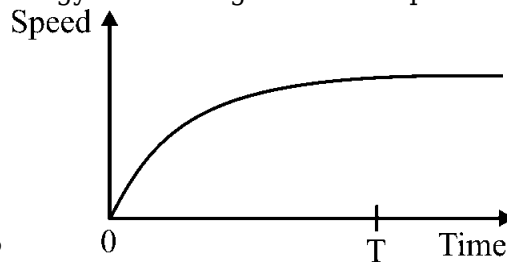
(A) The driver should steer the boat at an angle 30° with upstream.

(B) If time to cross river is 1 minute then width of river is $150\sqrt{3}$ m.

(C) Minimum time to cross the river is half of the actual time to cross the river.

(D) Shortest distance to cross the river is half of the actual distance covered by the boat.

2) The variation of speed with time for a ball falling vertical in air is shown below. During the time from 0 to T, the ball gains kinetic energy and loses gravitational potential energy ΔE_g . Which of the



following statement(s) is/are true?

- (A) ΔE_g is equal to the gain in kinetic energy
- (B) ΔE_g is greater than the gain in kinetic energy
- (C) ΔE_g is greater than the work done against air resistance
- (D) Air friction initially increases and then becomes constant.

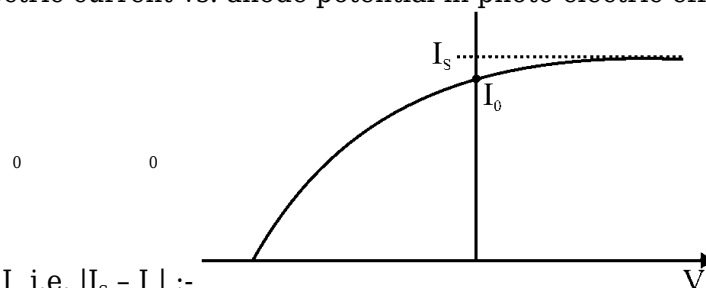
3) The coefficient of friction between the wheels and the ground is $\mu = 0.1$. The maximum speed with which cyclist can drive without slipping is 10 m/s. The strength of the air resistance acting on the bicyclist is proportional to the square of its velocity and is given by $f = -\alpha v^2$. Mass of cyclist with bicycle is $M = 100$ kg.

- (A) The value of α is $2 \text{ N s}^2/\text{m}^2$.
- (B) The value of α is $1 \text{ N s}^2/\text{m}^2$.
- (C) static friction of 25 N acts on the cyclist when he moves with 5 m/s.
- (D) kinetic friction of 25 N acts on the cyclist when he moves with 5 m/s.

4) A particle of mass '4m', kinetic energy = E, strikes a free neutral H atom in its ground state, at rest :-

- (A) For any excitation to happen in H-atom, E may be 20.4 eV.
- (B) For excitation to happen, E may be 60 eV.
- (C) If the striking particle was having more mass, it could have caused excitation through lesser energy.
- (D) If the striking particle was having more mass, it could have caused excitation through more energy.

5) The graph shows photoelectric current vs. anode potential in photo electric effect set up. The

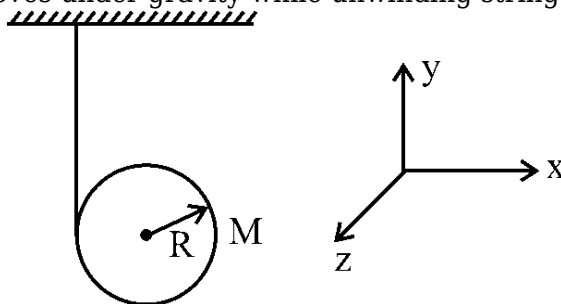


difference of currents I_s and I i.e. $|I_s - I|$:-

- (A) Depends upon geometry (shape, size & orientation) of anode and cathode.
- (B) Does not depend upon geometry of anode and cathode.

- (C) Can not become exactly zero.
 (D) Must depend on number of photons striking the cathode.

6) A disc of mass M radius R moves under gravity while unwinding string under non slip condition.

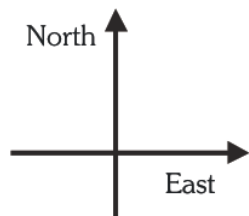


Mark the correct statement(s):

- (A) Acceleration of the disc is $\frac{2g}{3}$
 (B) Rate of $(\vec{L})$ change of angular momentum of disc $\frac{d\vec{L}}{dt}$ about a point along string is in $-\hat{k}$ direction.
 (C) Angular momentum of disc is conserved about centre of mass.
 (D) Rate of change of angular momentum $\frac{d\vec{L}}{dt}$ of disc about center of mass is in $+\hat{k}$ direction.

SECTION-I (ii)

1) A boat is being rowed in a river in presence of wind that is blowing uniformly from south-west. River currents have velocity towards the east and boat is moving towards the north. Ignore the effect of wind on boat's motion.



List-I

- (P) Direction in which boat is being steered
 (Q) Direction in which a flag on the boat may flutter
 (R) Direction of velocity of water relative to boat
 (S) Direction of velocity of air relative to a piece of wood floating on river.

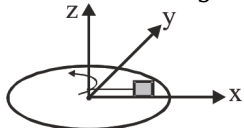
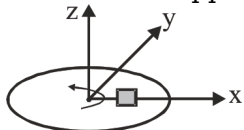
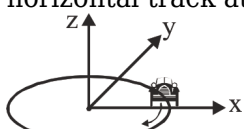
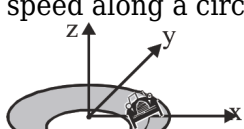
List-II

- (1)
- (2)
- (3)
- (4)
- (5)

- (A) $P \rightarrow 1; Q \rightarrow 2,4; R \rightarrow 4; S \rightarrow 1,3$
 (B) $P \rightarrow 2; Q \rightarrow 1,3; R \rightarrow 2,4; S \rightarrow 3$
 (C) $P \rightarrow 3,2; Q \rightarrow 1,4; R \rightarrow 3; S \rightarrow 2,4$

(D) $P \rightarrow 1,4; Q \rightarrow 1,3; R \rightarrow 2; S \rightarrow 1$

2) Mark the matching options for situations shown in respective figures of list-I at the instant, when particle / car is located on the x-axis as shown :-

List-I		List-II	
(P)	Block attached to string is moving along a circle on rough surface 	(1)	Force due to friction may have non-zero x-component
(Q)	Block is placed on a disc rotating with non-uniform angular velocity as shown below. There is no slipping between block & disc. 	(2)	Force due to friction may have non-zero y-component.
(R)	Car is moving on ground along a circular horizontal track at a constant speed 	(3)	Force due to friction may have non-zero z-component.
(S)	Car is moving on ground at constant speed along a circular banked track. 	(4)	Force due to friction may be zero.
		(5)	Centripetal force is only due to friction.

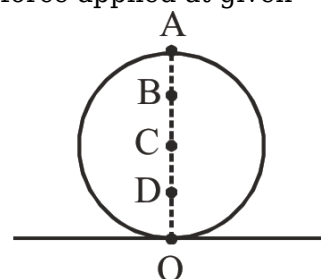
(A) $P \rightarrow 2; Q \rightarrow 1,2,5; R \rightarrow 1,5; S \rightarrow 1,3$

(B) $P \rightarrow 2; Q \rightarrow 1,2; R \rightarrow 3,4; S \rightarrow 5$

(C) $P \rightarrow 1,2,3; Q \rightarrow 2,3,4; R \rightarrow 4,5; S \rightarrow 4,5$

(D) $P \rightarrow 2; Q \rightarrow 1,2,5; R \rightarrow 1,2; S \rightarrow 2,4$

3) A uniform solid cylinder is placed on a rough horizontal surface. Two forces $|\vec{F}_1| = F$ and $|\vec{F}_2| = 2F$ are applied simultaneously on the cylinder at the given points in the same direction. Given C is the centre of mass and $AB = DO = R/2$. Match the List-I mentioning the force applied at given



points with List-II mentioning the direction of frictional force at that instant.

List-I		List-II	
(P)	F_2 at A and F_1 at C	(1)	Forward
(Q)	F_1 at B and F_2 at D	(2)	Backward
(R)	F_1 at A and F_2 at D	(3)	Zero
(S)	F_2 at A and F_1 at D	(4)	Cannot be determined
		(5)	None of these

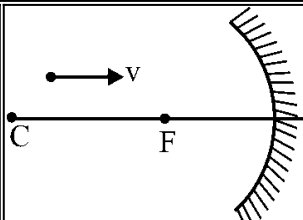
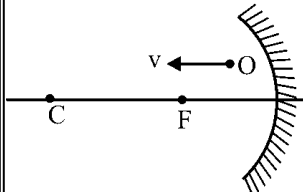
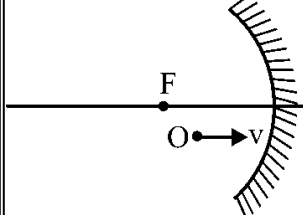
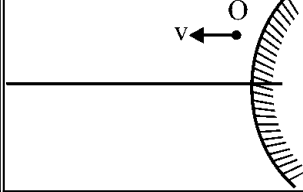
(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

4) In list-I possible instantaneous velocity vector of the **object** with respect to ground are shown. The corresponding velocity vectors of images are the situations shown in list-II. Match list-I with the list-II.

List-I		List-II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	
		(5)	

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

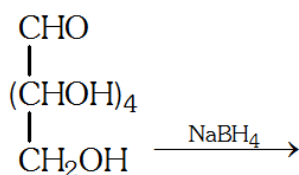
(B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

PART 2 : CHEMISTRY

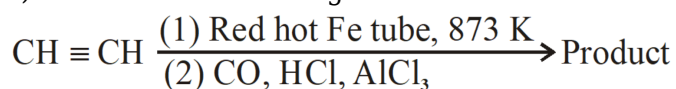
SECTION-II



1) Glucose

How many optically active isomers of product are possible ?

2) Consider the following chemical reaction.



The number of sp^2 hybridized carbon atom(s) present in the product is _____.

3) A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0 % to the total weight of the hydrolyzed products. The number of glycine units present in the decapeptide is

4) No. of S - S linkage in $\text{S}_3\text{O}_9 = m_1$

No. of P - P linkage in $(\text{HPO}_3)_3 = m_2$

No. of S - S linkage in $\text{S}_2\text{O}_8^{2-} = m_3$

No. of O - O linkage in $\text{S}_2\text{O}_6^{2-} = m_4$

Find out $m_1 + m_2 + m_3 + m_4$

5) How many compounds will give positive iodoform test but negative Tollen's test?

(a)	CH_3CHO	(b)	$\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\parallel}}\text{C}-\text{CH}_3$
(c)	$\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CH}_3$	(d)	
(e)		(f)	
(g)	$\text{CH}_3-\text{CH}_2-\text{OH}$	(h)	CH_3-COOH
(i)	PhCOCH_3		

6) Consider the following molecules/ion calculate (x . y + z)

CO, $\text{CN}^\ominus$ and N_2

here, x = sum of total number of σ bonds in given three species

y = sum of total number of π bonds in given three species

z = sum of total number of lone pairs in given three species

7) For electron in hydrogen atom wave function is given :

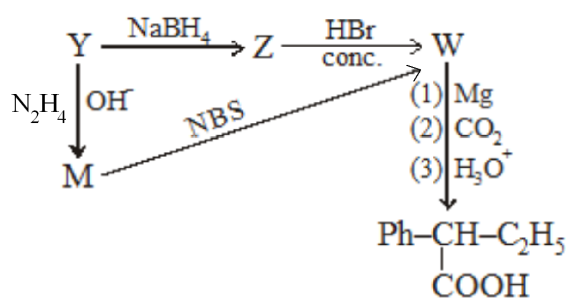
$$\psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-\frac{r}{a_0}}$$

Find the ratio of probability density of electron at ($r = a_0$) to the probability density of electron at nucleus ($r = 0$).

[Given : $\ln \left(\frac{1}{0.14} \right) = 2$]

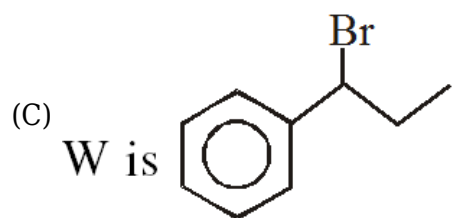
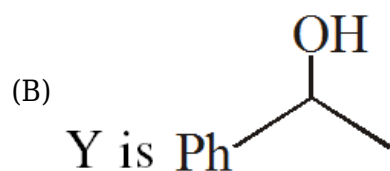
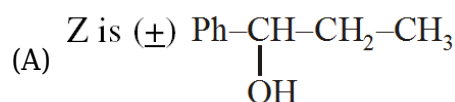
8) The elevation in boiling point, when 13.44 g of freshly prepared CuCl_2 are added to one kilogram of water, is [Some useful data, $K_b(\text{H}_2\text{O}) = 0.52 \text{ kg K mol}^{-1}$, mol. wt. of $\text{CuCl}_2 = 134.4 \text{ gm}$]

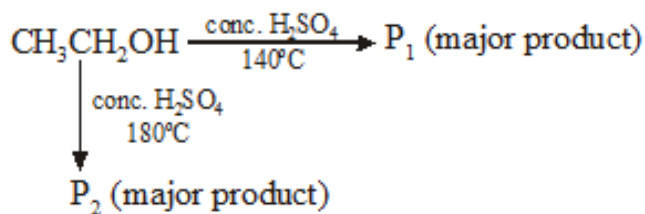
SECTION-I (i)



1)

Product in a given conversion are :





Correct statements :

- (A) P_1 & P_2 both are Diethyl ether
- (B) P_1 is diethyl ether
- (C) P_2 is ethene
- (D) P_1 & P_2 both are ethene

3)

An electrochemical cell is made by combining two half cells

(X) $\text{AgNO}_3(0.05 \text{ M}) \mid \text{Ag(s)}$ and

(Y) $\text{AgNO}_3(0.01 \text{ M}) \mid \text{Ag(s)}$

Choose the correct statement

- (A) The E_{cell} can be increased if concentration of AgNO_3 in half cell X is increased
- (B) The E_{cell} will increase if concentration of AgNO_3 in cell Y is increased
- (C) The E_{cell} will increase if concentration of AgNO_3 in cell X is decreased
- (D) The E_{cell} will increase if concentration of AgNO_3 in cell Y is decreased

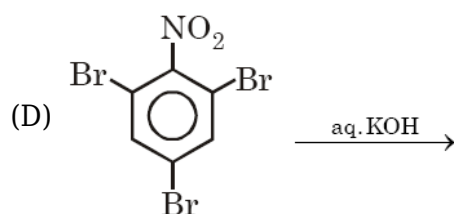
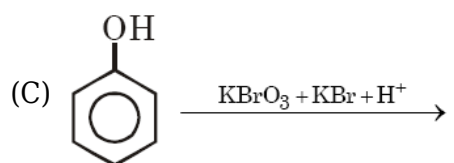
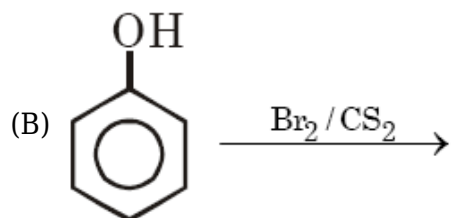
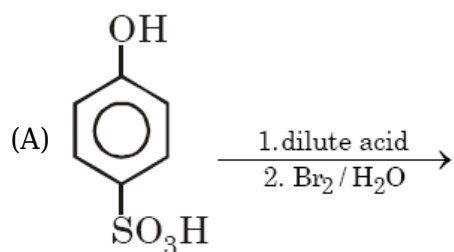
4) 1 mol of NH_3 gas ($\gamma = 1.33$) at 27°C is expanded reversibly under adiabatic conditions to increase volume 8 times the initial volume. Correct option(s) is/are

- (A) final temp. = 150 K
- (B) work done = -400 cal
- (C) final temp. = 250 K
- (D) work done = -900 cal

5) Which of the following pairs produce same gas?

- (A) $(\text{Ca}_3\text{N}_2 + \text{dil. HCl})$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
- (B) $[(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \text{ on heating}]$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
- (C) $(\text{NH}_4\text{NO}_3 \text{ on heating})$ and $[\text{Hg}(\text{NO}_3)_2 \text{ on heating}]$
- (D) $(\text{NH}_4\text{Cl on heating})$ and $[\text{NaNO}_3 + \text{Zn} + (\text{aq.}) \text{NaOH}_{(\text{aq})} \text{ on heating}]$

6) Which of the following reaction give tribromo phenol as product?



SECTION-I (ii)

1)

List-I (Reactions)		List-II (Observations)	
(P)	CrO_5 (on standing in acidic medium)	(1)	Redox reaction
(Q)	$\text{Cr}_2\text{O}_7^{2-} + \text{C}_2\text{O}_4^{2-} \xrightarrow{\text{H}^+}$	(2)	Evolution of O_2 is associated
(R)	$\text{MnO}_2 + \text{conc. KOH} + \text{O}_2 \xrightarrow{\Delta}$	(3)	Green colour solution/compound is formed as product
(S)	$\text{KMnO}_4 + \text{conc. KOH} \xrightarrow{\Delta}$	(4)	Orange/Yellow colour solution/compound is formed as product
		(5)	Decomposition reaction

The correct option is :-

(A) P $\rightarrow$ 3,4,5; Q $\rightarrow$ 1,4; R $\rightarrow$ 2,4; S $\rightarrow$ 1,2,3,4

(B) P $\rightarrow$ 1,2; Q $\rightarrow$ 2,4; R $\rightarrow$ 1,2,3; S $\rightarrow$ 1,2,4

(C) P $\rightarrow$ 1,2,3; Q $\rightarrow$ 1,3; R $\rightarrow$ 1,3; S $\rightarrow$ 1,2,3

(D) P $\rightarrow$ 3,4; Q $\rightarrow$ 1,2,3; R $\rightarrow$ 2,3,5; S $\rightarrow$ 2,3,4

2) Match the List-I with List-II.

List-I (Oxidizing agent)		List-II (Characteristics)	
(P)	H ₂ O ₂	(1)	Colored product on reaction with NO ₂ ⁻
(Q)	Cl ₂ water	(2)	Precipitate/turbidity formation on reaction with H ₂ S
(R)	Acidified KMnO ₄	(3)	Colored product on reaction with Br ⁻
(S)	Acidified K ₂ Cr ₂ O ₇	(4)	Gas formation on reaction with F ⁻
		(5)	Only colorless products on reaction with SO ₃ ²⁻

The correct option is :-

- (A) P → 1,2,3,5; Q → 1,2,3,5; R → 1,2,3,5; S → 1,2,3,5
 (B) P → 1,2,3,5; Q → 1,3,5; R → 1,4,5; S → 2,5
 (C) P → 2,3; Q → 2,3,5; R → 1,3; S → 1,2,3
 (D) P → 2,3; Q → 2,5; R → 3,5; S → 3

3) Match the half-life in List-I with the order in List-II for the reaction :

A → Products

	List-I		List-II
(I)	t _{1/2} = constant	(P)	First order
(II)	t _{1/2} ∝ a	(Q)	Third order
(III)	t _{1/2} ∝ 1/a	(R)	Second order
(IV)	t _{1/2} ∝ 1/a ²	(S)	Zero order
		(T)	Half life increases with concentration

where a = Initial concentration of the reactant

- (A) I → PQ; II → RS; III → RT; IV → S
 (B) I → Q; II → T; III → RS; IV → T
 (C) I → P; II → ST; III → R; IV → Q
 (D) I → R; II → S; III → P; IV → Q

4) Answer the following by appropriately matching the lists based on the information given in the paragraph.

	List-I		List-II
(I)	o-nitrophenol	(P)	Intermolecular H-bonding
(II)	Chloral hydrate	(Q)	Intramolecular H bonding
(III)	p-hydroxybenzaldehyde	(R)	Higher vapour pressure than its o-/p- form, if they exist for the molecule
(IV)	Acetic acid	(S)	Chlorine forms H-bond in this compound

(T)	Molecular mass is found twice than ideal experimentally
-----	---

Which of the following options has the **INCORRECT** combination considering List-I and List-II ?

- (A) I → R; II → Q; III → P; IV → T
 (B) I → Q; II → R; III → P; IV → T
 (C) I → R; II → P; III → R; IV → T
 (D) I → R; II → R; III → P; IV → P

PART 3 : MATHEMATICS

SECTION-II

1) Let $S_n = \frac{1}{1^n} + \frac{1}{3^n} + \frac{1}{5^n} + \dots \infty$ and $\int_0^1 \frac{x(\ln x)^2}{1-x^4} dx = mS_3$ then value **m** is

2)
$$\int_0^1 \left(\sum_{k=1}^{100} \frac{x}{x^2 + k^2} \right) \cdot \left(\prod_{k=1}^{100} (x^2 + k^2) \right) dx = \frac{1}{\lambda} \left[\prod_{k=1}^{100} (1^2 + k^2) - (100)^\lambda \right]$$
 then value of $\frac{1}{\lambda}$ is equal to

3) Let $f(x)$ and $g(x)$ are two quadratic functions satisfying $g(x) = -f(4 - x) \forall x \in \mathbb{R}$ and graph of $g(x)$ passes through the vertex of graph of $f(x)$. The four x -intercepts on the two graphs have x -coordinates x_1, x_2, x_3 and x_4 in increasing order and $x_3 - x_2 = 6$. If value of $x_4 - x_1$ is $m + n\sqrt{p}$, where m, n, p are positive integers and p is not divisible by the square of any prime, then $\left(\frac{mp}{n}\right)$ is equal to

4) Consider a system of linear equations
$$\begin{bmatrix} \sin \alpha & \cos \alpha & 0 \\ \cos \alpha & 0 & \sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha = \frac{n\pi}{2}, n \in \mathbb{I},$$
 then the value of $(x - y)^2 + 4z^2$ is

5) Given $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ and $\int_0^1 \frac{\log_e x}{1-x^2} dx = -\frac{\pi^2}{\lambda}$, then the value of λ equals

6) If $f : A \rightarrow B$ be an injective (one-one) and subjective (onto) function such that $4^{f(x)} + 2^{-x} = 8$. If smallest integer in set A is m and largest integer in set B is n then $\left\lfloor \frac{n}{m} \right\rfloor$ is

7) Evaluate the value of the limit
$$L = \lim_{x \rightarrow 0} \left(\frac{1}{\ln(x + \sqrt{x^2 + 1})} - \frac{1}{\ln(x + 1)} \right)$$

8)
$$\lim_{n \rightarrow \infty} \frac{\pi}{2n} \left(1 + \cos \frac{\pi}{2n} + \cos \frac{2\pi}{2n} + \dots + \cos \frac{(n-1)\pi}{2n} \right)$$
 equal to

SECTION-I (i)

1) The number of values of x satisfying the equation

$$2^{\ln \cos \csc x} + \log_2 \left(\frac{1 - \sin x}{\cos x} \right) + \log_4 \left((1 + \sin^2 x + 2 \sin x) \cdot \sec^2 x \right) = (\sin x + \csc x)^{\ln \csc x}$$
 where x defined is equal to -

(A) The value of $\log_{\sqrt{5}} (\sqrt{2} + 1) + \log_{\sqrt{5}} \sqrt{3 - 2\sqrt{2}}$

(B) The value of $\sum_{r=1}^{89} \log_{\sin 20^\circ} (\tan r^\circ)$

(C) The value of $\log_{\sin 20^\circ} \csc 20^\circ$

(D) The value of $\sum_{r=1}^6 \cos \left(\frac{r\pi}{7} \right)$

2) Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a strictly increasing function such that $x \cdot f(x) + 1 > 0$ and

$$f(x) \left[f \left(f(x) + \frac{1}{x} \right) \right] = 1$$
 for all $x > 0$, then which of the following is/are correct-

(A) $f(1) = 2 \sin \left(\frac{\pi}{10} \right)$

(B) $f(1) = 2 \sin \left(\frac{11\pi}{10} \right)$

(C) $\operatorname{sgn} f(x) = 1$ for $x > 0$

(D) $\operatorname{sgn} f(x) = -1$ for $x > 0$

3) If $A = \sum_{n=1}^8 \sqrt{3 + \sqrt{n}}$ and $B = \sum_{n=1}^8 \sqrt{3 - \sqrt{n}}$, then which of the following is/are not correct ?

(A) $\frac{A}{B} > \frac{9}{4}$

(B) $\frac{A}{B} > \frac{5}{2}$

(C) $\frac{A}{B} > 3$

(D) $\frac{A}{B} > \frac{13}{4}$

4) The roots of equation $ax^2 + bx + c = 0$ are $\alpha = z^2 + z^4 + \dots + z^{2n}$ and $\beta = z + z^3 + \dots + z^{2n-1}$, where $z = e^{i\left(\frac{2\pi}{2n+1}\right)}$, $n \in \mathbb{N}$, then

(A) $\alpha + \beta = -1$

(B) $\alpha + \beta = 1$

(C) given equation can be written as $x^2 + x + \frac{1}{4}\sec^2\left(\frac{\pi}{2n+1}\right) = 0$

(D) equation can be written as $x^2 + x + \frac{1}{4}\cos^2\left(\frac{\pi}{2n+1}\right) = 0$

5) Let for $x \in [1, \infty)$, $\int_1^x f(x) dx = \frac{x^2+1}{2x} \left(f(x) - x + \frac{2x}{x^2+1} \right) - \log_e 2$, then for $x \in [1, \infty)$

(A) range of $g(x) = \frac{xf'(x) - f(x)}{x}$, where $x \in \mathbb{R}$ is $[0, 2)$

(B) $\lim_{x \rightarrow 2} \left[\frac{f(x)}{x} \right] = 1$ (where $[.]$ denotes greatest integer functions)

(C) there exist atleast one $x = \alpha$ such that $f'(x) - f(x) > 0 \forall x > \alpha, \alpha \in (1, \infty)$

(D) there exist atleast one $x = \alpha$ such that $f'(x) - f(x) < 0 \forall x > \alpha, \alpha \in (1, \infty)$

6) Let $f: [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that for all non-negative integer n , $\int_0^1 x^n f(x) dx = 0$. Then

(A) $f(0) = 0$

(B) $f(x)$ must be identically zero

(C) $f(1) = 1$

(D) $\exists x_1, x_2 \in [0, 1]$ such that $f(x_1)f(x_2) < 0$

SECTION-I (ii)

1) Match the following List-I with List-II

List-I		List-II	
(P)	$\sum_{k=1}^{\infty} \frac{1}{2^k} \tan \frac{\pi}{2^{k+2}}$	(1)	$\ln 2 - \frac{1}{2}$
(Q)	$\sum_{k=1}^{\infty} \cot^{-1} \left(k^2 + \frac{3}{4} \right)$	(2)	$\frac{\pi}{2}$

(R)	$\sum_{k=1}^{\infty} \frac{1}{(2k-1)2k(2k+1)}$	(3)	$\frac{1}{2} - \frac{\pi}{8}$
(S)	$\sum_{k=1}^{\infty} \frac{1}{(4k+1)(4k-1)}$	(4)	$\frac{4}{\pi} - 1$
		(5)	$\tan^{-1} 2$

The correct option is:

- (A) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 3$
 (B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 3$
 (C) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$
 (D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 1$

$$f(x) = \begin{cases} 2ax - 1; & x \in [0, 1] \\ 1 - bx; & x \in (1, 2] \\ cx - 5; & x \in (2, 3] \\ 1 - dx; & x \in (3, 4] \\ ex - 19; & x \in (4, 5] \\ 6; & x \geq 5 \end{cases}$$

2) Let be a continuous function $\forall x \in \mathbb{R}$
 Match the following List.

List-I		List-II	
(P)	$\lim_{n \rightarrow \infty} \sum_{r=1}^n \sin\left(\frac{r}{n^2}\right)$	(1)	a
(Q)	If $[.]$ denotes the greatest integer function, then $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left[\frac{n}{2^r} + \frac{1}{2} \right]$	(2)	b
(R)	$2 \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sqrt{1 + \frac{r}{n^2}} - 1 \right)$	(3)	c
(S)	$\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n^2 + n + 2r} \right)$	(4)	d
		(5)	e

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 2$
 (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$
 (C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 5$
 (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 1$

$$f(x) = \begin{cases} x|x| \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0, \end{cases} \text{ and}$$

3) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$g(x) = \begin{cases} 1 - 2x, & 0 \leq x \leq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Let $a, b, c, d \in \mathbb{R}$. Define the function $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = a f(x) + b \left(g(x) + g\left(\frac{1}{2} - x\right) \right) + c(x - g(x)) + d g(x), \quad x \in \mathbb{R}$$

Match each entry in **List-I** to the correct entry in **List-II**.

List-I		List-II	
(P)	If $a = 0, b = 1, c = 0$ and $d = 0$, then	(1)	h is one-one.
(Q)	If $a = 1, b = 0, c = 0$ and $d = 0$, then	(2)	h is onto.
(R)	If $a = 0, b = 0, c = 1$ and $d = 0$, then	(3)	h is differentiable on $\mathbb{R}$.
(S)	If $a = 0, b = 0, c = 0$ and $d = 1$, then	(4)	the range of h is $[0, 1]$.
		(5)	the range of h is $\{0, 1\}$.

The correct options is

- (A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
 (B) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$
 (C) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$
 (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

4) Match the following :

List-I		List-II	
(P)	$I = \left[\int_0^\infty \left[\frac{e}{e^x} \right] dx \right] =$ (where $[x]$ denotes the greatest integer less than or equal to x)	(1)	0
(Q)	The numbers 1, 2, 3,9 are divided into 3 groups each containing 3 digits. P is the probability that atleast in one group, the product of 3 digits exceeds 71 then the value of $3P$ is	(2)	1
(R)	The number of non negative solutions of the equation $ \tan^{-1} x = \sqrt{(x^2 + 1)^2 - 4x^2}$ is -	(3)	2
(S)	Consider the function $f(x) = \begin{cases} \sqrt{[x] + \sqrt{\{x\}}; & x \geq 0 \\ \sin \pi [x] ; & x < 0 \end{cases}$ where $[.]$ and $\{.\}$ denotes greatest integer function and fraction part of function respectively. Number of points of discontinuity is	(4)	3
		(5)	4

- (A) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

ANSWER KEYS

PART 1 : PHYSICS

SECTION-II

Q.	1	2	3	4	5	6	7	8
A.	1.73	8.80	12.50	4.50	0.72	1.33	2.14	12.50

SECTION-I (i)

Q.	9	10	11	12	13	14
A.	A,B,C	B,C,D	B,C	B,D	A,C	A,B

SECTION-I (ii)

Q.	15	16	17	18
A.	A	A	C	D

PART 2 : CHEMISTRY

SECTION-II

Q.	19	20	21	22	23	24	25	26
A.	8.00	7.00	6.00	0.00	5.00	24.00	0.14	0.16

SECTION-I (i)

Q.	27	28	29	30	31	32
A.	A,C,D	B,C	A,D	A,D	B,D	A,C

SECTION-I (ii)

Q.	33	34	35	36
A.	C	C	C	C

PART 3 : MATHEMATICS

SECTION-II

Q.	37	38	39	40	41	42	43	44
A.	0.25	0.50	3.00	1.00	8.00	0.50	-0.50	1.00

SECTION-I (i)

Q.	45	46	47	48	49	50
A.	A,B,D	B,D	B,C,D	A,C	A,B,D	A,B

SECTION-I (ii)

Q.	51	52	53	54
A.	A	B	C	D

SOLUTIONS

PART 1 : PHYSICS

1)

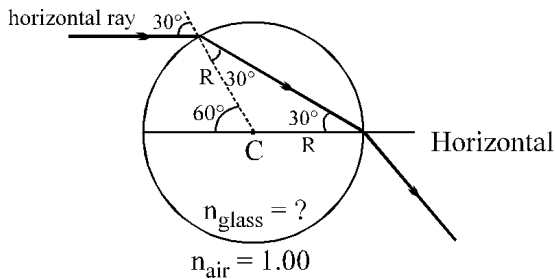
Explain Question: Question asks to find refractive index of sphere.

Concept: Snell's Law, $n_1 \sin i = n_2 \sin r$

Solution:

$$1 \times \sin 60^\circ = \mu \sin 30^\circ$$

$$\mu = \sqrt{3} = 1.73$$



Final Answer: 1.73

2)

Explain Question: Question is asking to find natural length of spring.

Concept: $F_{\text{net}} = F_C = \frac{mv^2}{r}$

Solution:

$$(20 - x_0)k - (16 - x_0)k$$

$$= m \times 400 \times 20$$

$$k \times (16 - x_0) = m \times 400 \times 36$$

$$4k = 8000 \text{ m}$$

$$2000 \text{ m} (16 - x_0) = 14400 \text{ m}$$

$$x_0 = 16.0 - 7.2 = 8.80$$

Final Answer: 8.80

3)

Explain Question: Question asks to find average power delivered by the engine.

Concept: $\vec{P} = \vec{F} \cdot \vec{V}$ $\langle P \rangle = F \langle V \rangle$

Solution:

$$P_{\text{avg}} = \frac{W}{t} = \frac{5 \times 2}{\sqrt{\frac{2 \times 2}{25/4}}} = 12.5$$

Final Answer: 12.50

4)

Explain Question: Question asks to find the location where two bright fringe of

bichromatic light in YDSE coincide.

Concept: $B = \frac{n\lambda}{d}$

Solution:

If n^{th} bright of 1st light and n^{th} bright of 2nd light coincide.

$$m\lambda_1 = n\lambda_2$$

$$m \times 750 = n \times 900$$

$$\Rightarrow \frac{m}{n} = \frac{6}{5}$$

Just after centre $\Rightarrow m = 6, n = 5$

Minimum distance from centre

$$= \frac{m\lambda_1 D}{d} = \frac{6 \times 750 \times 10^{-9} \times 2}{2 \times 10^{-3}} = 4.5 \text{ mm}$$

Final Answer: 4.50

5)

Explain Question: Question asks to find the ratio of frequency of photon transition to that of orbital motion frequency in a Bohr hydrogen atom.

Concept: $f \propto \frac{2^2}{n^3}$ and $f = \frac{\Delta E}{h}$

Solution:

$$E_n = -\frac{mz^2e^4}{8\varepsilon_0^2n^2h^2}$$

$$\text{so, } h\nu = +\frac{mz^2e^4}{8\varepsilon_0^2h^2} \left[\frac{1}{16} - \frac{1}{25} \right]$$

$$\square \quad \nu = \frac{mz^2e^4}{8\varepsilon_0^2h^3} \left[\frac{9}{16 \times 25} \right] \dots(1)$$

$$\text{and frequency } \nu_4 = \frac{1}{T} = \frac{\nu}{2\pi r}$$

$$= \left(\frac{ze^2}{2\varepsilon_0nh} \right) \frac{1}{2\pi} \left(\frac{\pi mze^2}{\varepsilon_0h^2n^2} \right) = \frac{z^2e^4m}{4\varepsilon_0^2n^3h^3} \dots(2)$$

$$\square \quad \nu/\nu_4 = 18/25 = 0.72$$

Final Answer: 0.72

6)

Explain Question: The question asks to find potential difference between two points in time varying magnetic field.

Concept: $\varepsilon = \frac{-\Delta B}{dt}$

Solution:

The direction of induced electric field is circular and hence potential drop across wire OA and OB is zero. Hence all induced emf will be in the wire AB.

$$e = -A \frac{dB}{dt} = - \left(-\frac{\pi \times 1^2}{6} \times \frac{6}{\pi} \times 2 \cos 2t \right) = 2 \cos 2t$$

$$\square e_{\max} = 2 \text{ volt}$$

$$V = e - i(r/3)$$

Final Answer: 1.33

7)

Explain Question: The question asks to find rms current in an AC circuit.

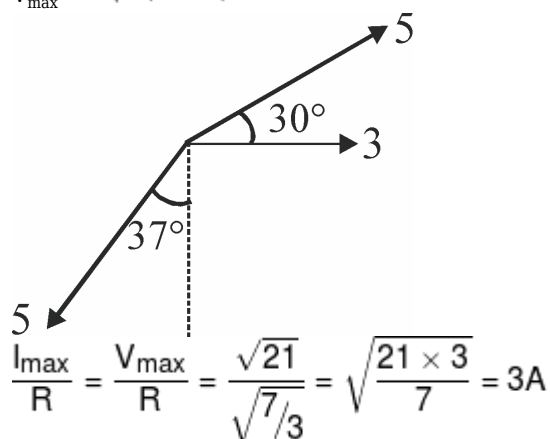
Concept: $i_{\text{rms}} = \frac{V_{\text{rms}}}{R}$

Solution:

$$v_1 = 3 \sin \omega t; v_2 = 5 \sin(\omega t + \phi_1);$$

$$v_3 = 5 \sin(\omega t - \phi_2)$$

$$V_{\max} = \sqrt{\left(\frac{5\sqrt{3}}{2}\right)^2 + (1.5)^2} = \sqrt{21}$$



$$I_{\text{rms}} = \frac{3}{1.4} = 2.14$$

Final Answer: 2.14

8)

Explain Question: Question asks to find energy in inductor in a LC circuit

Concept: Energy oscillates between capacitor and inductor.

$$W_{\text{battery}} = \frac{1}{2} Li^2 + \frac{q^2}{2C}$$

Solution:

For $I_{\max}$ in inductor, $\frac{dI}{dt} = 0$

Hence $q = CE$

$$W_{\text{battery}} = \frac{q^2}{2C} + \frac{1}{2} Li^2$$

Energy stored inside inductor

$$\frac{1}{2} Li^2 = \frac{1}{2} CE^2 = 12.50 \text{ J}$$

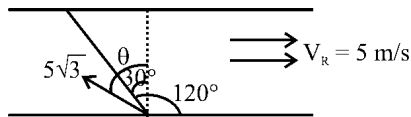
Final Answer: 12.50

9)

Explain Question: Question asks to find different parameters of a motion of boat in flowing rivers.

Concept: $\vec{V}_{br} = \vec{V}_b - \vec{V}_r$

Solution:



$$\begin{array}{c} \uparrow 5\sqrt{3} \cos \theta \\ \leftarrow 5\sqrt{3} \sin \theta \quad \rightarrow V_r \\ \end{array} = \begin{array}{c} \uparrow 5\sqrt{3} \cos \theta \\ \leftarrow 5\sqrt{3} \sin \theta \quad \rightarrow \end{array}$$

$$\tan 30 = \frac{5\sqrt{3} \sin \theta - 5}{5\sqrt{3} \cos \theta}$$

$$5\sqrt{3} \cos \theta = 15 \sin \theta - 5\sqrt{3} \Rightarrow \theta = 60^\circ$$

Hence angle from up stream = 30°

$$\text{width} = 5\sqrt{3} \cos \theta \times 60 = 150\sqrt{3} \text{ m}$$

$$t_{\min} = \frac{w}{5\sqrt{3}}$$

$$t_{\text{actual}} = \frac{w}{5\sqrt{3} \cos \theta} \Rightarrow \frac{t_{\min}}{t_{\text{actual}}} = \cos \theta = \frac{1}{2}$$

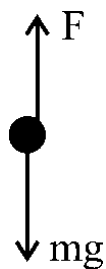
Final Answer: Option (A, B, C) is Correct.

10)

Explain Question: The question asks to moment on energy losses and gain of a falling body in a viscous fluid.

Concept: $F_v = 6\pi\eta r v$ (Stoke's law)

Solution:



Acceleration first decreases and then becomes zero.

Final Answer: Option (B,C,D) is Correct.

11)

Explain Question: The question asks to find acceleration of cyclistist under friction and drag force.

Concept: $\mu mg = \alpha v^2$

Solution:

$$\mu mg = \alpha v^2$$

$$0.1 \times 100 \times 10 = \alpha (10)^2$$

$$\square a = 1 \text{ N s}^2/\text{m}^2$$

Final Answer: Option (B,C) is Correct.

12)

Explain Question: Question asks to find excitation energy of collision of atoms.

Concept: $\text{DKE} = \Delta \text{KE} = \frac{1}{2} m v_{\text{net}}^2$

$\Delta E_{\text{excitation}} = \text{Loss in KE.}$

Solution:

$$\Delta \text{KE}_{\text{max}} = \frac{1}{2} \mu V_r^2$$

$$= \frac{1}{2} \frac{M(4m)}{5m} V^2 = \Delta E$$

$$\frac{1}{2} (4m) V^2 = 5 \Delta E$$

for excitation, $\Delta E = 10.2 \text{ eV min.}$

$$5 \Delta E = 51 \text{ eV}$$

Final Answer: Option (B, D) is Correct.

13)

Explain Question: Question asks to find different parameters of photoelectric affect.

Concept: No. of electrons having the plane will depend on distance shape & size.

Solution:

No. of electrons reaching the plate will depend on distance, shape and size.

Final Answer: Option (A,C) is Correct.

14)

Explain Question: Question asks to find acceleration, and angular momentum of a disc under CTRM.

Concept: $\vec{\tau} = I \vec{\alpha}$; $\vec{\tau} = \frac{d\vec{L}}{dt}$

Solution:

(A) $mg - T = ma$

$$T.R = \frac{MR^2}{2} \alpha$$

$$a = R\alpha$$

(C) angular momentum about centre of mass not conserve because tension force provide torque

(D) $\frac{d\vec{L}}{dt} = \text{torque} = \vec{r} \times \vec{F}$

about centre of mass

$$\vec{r} \Rightarrow -\hat{i}$$

$$\vec{F} \Rightarrow -\hat{j} \quad (\text{tension force})$$

$$\vec{\tau} = -(\hat{i} \times \hat{j}) = -\hat{k}$$

Final Answer: Option (A, B) is Correct.

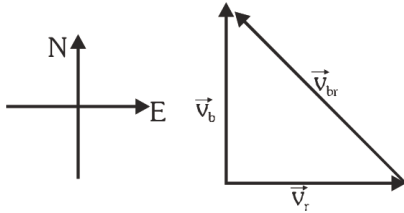
15)

Explain Question: Question asks to find direction of relative velocities of boat wrt river and flag wrt wind.

Concept: $\vec{v}_{br} = \vec{v}_b - \vec{v}_r$

Solution:

(A) $\vec{v}_b = \vec{v}_{br} + \vec{v}_r$

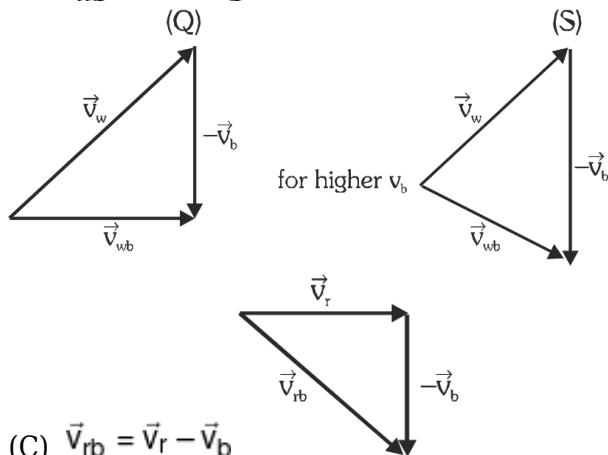


[Actual = Relative + Reference]

(B) Flag flutters in direction

$\vec{v}_{wb}$ (velocity of wind w.r.t. boat)

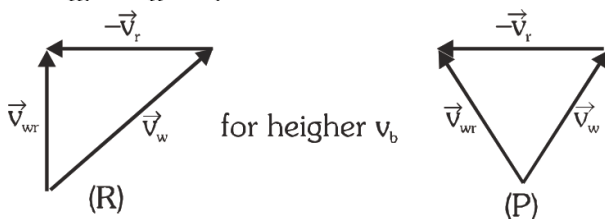
$\vec{v}_{wb} = \vec{v}_w - \vec{v}_b$



(C) $\vec{v}_{rb} = \vec{v}_r - \vec{v}_b$

(D) Wood floating on river moves with $\vec{v}_r$

$\vec{v}_{wr} = \vec{v}_w - \vec{v}_r$



Final Answer: Option (A) is Correct.

16)

Explain Question: Question asks to find direction of frictional force on block in circular motion.

Concept: Frictional force may provide centripetal force in circular motion.

Solution:

(P) Friction along tangent only.

(Q) Friction provide centripetal and tangential acceleration.

(R) Friction provide only centripetal acceleration.

(S) If speed is not equal to $\sqrt{rg \tan \theta}$ then friction is along the incline of road to avoid slipping & if

speed is equal to $\sqrt{rg \tan \theta}$, no friction.

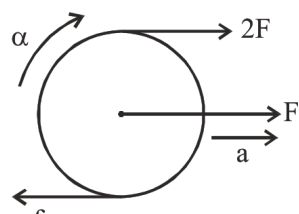
Final Answer: Option (A) is Correct.

17)

Explain Question: Question asks to find direction of frictional force in CTRM (pure rolling)

Concept: $\tau = I\alpha$; $F = ma$

Solution:



$$2F + F - f = ma \quad \dots (i)$$

$$fR + 2FR = \frac{mR^2}{2} \frac{a}{R} \quad \dots (ii)$$

from (i) & (ii)

$$f = -\frac{F}{3} \text{ or friction is in forward direction}$$

Similarly we can solve (Q, R, S)

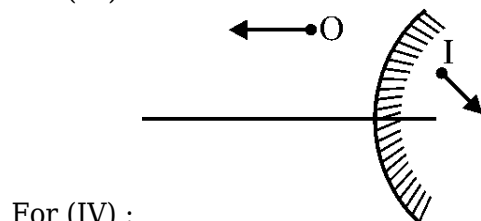
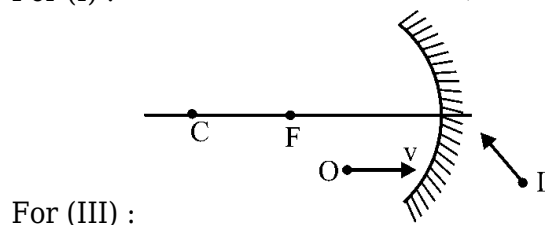
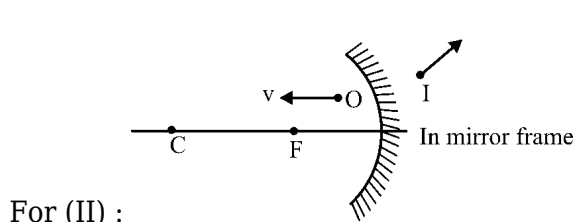
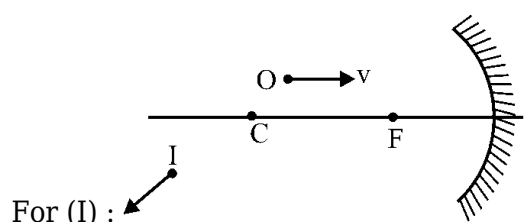
Final Answer: Option (C) is Correct.

18)

Explain Question: Question asks the direction of motion of image of an object in concave mirror

Concept: Theoretical

Solution:



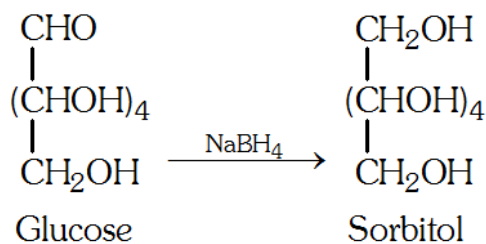
Final Answer: Option (D) is Correct.

PART 2 : CHEMISTRY

19) **Explain Question:** Identify the total number of chiral center in product formed in given reaction,

Concept: Apply concept of optical isomerism.

Solution:



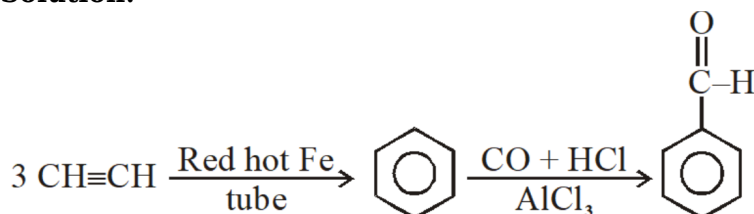
Total optically active isomer = $2^{n-1} = 2^{4-1} = 8$

Final Answer: (8.00)

20) **Explain Question:** Identify the final product of given reaction.

Concept: Preparation of benzene is followed by Gattermann - koch reaction

Solution:



In benzaldehyde total number of sp^2 'C' are 7.

Final Answer: (7.00)

21) **Explain Question:** Calculate number of glycine units present in the decapeptide

Concept: Number of glycine units = $\frac{\text{mass of hydrolysed products} \times \text{contribution of glycine}}{\text{molar mass of glycine}}$

Solution:

No. of peptide bonds = 9

Mass of hydrolysed products = $796 + 18 \times 9 = 958 \text{ g}$

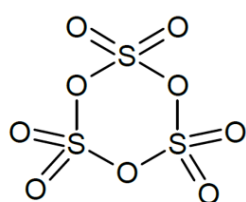
No. of glycine units = $958 \times \frac{47}{100} \times \frac{1}{75} = 6$

Final Answer: (6.00)

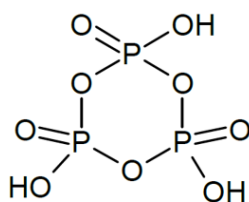
22) **Explain Question:** Calculate sum of $m_1 + m_2 + m_3 + m_4$

Concept: Structures of given oxides of different elements.

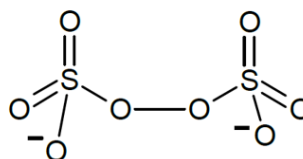
Solution:



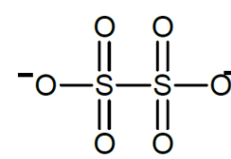
$m_1 = 0$



$(\text{HPO}_3)_3$
 $m_2 = 0$



$m_3 = 0$



$(\text{S}_2\text{O}_6^{2-})$
 $m_4 = 0$

Final Answer: (0.00)

23)

Question Explanation:

Count the number of compounds that test positive for iodoform but negative for Tollen's.

Concept: This question is based on iodoform and Tollen's tests.

Solution:

The iodoform test is positive for compounds containing a methyl ketone ($\text{CH}_3\text{CO}-$) or methyl carbinol ($\text{CH}_3\text{CHOH}-$) group.

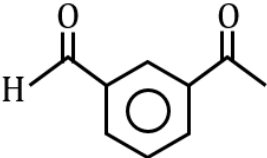
The Tollen's test is positive for aldehydes and alpha-hydroxy ketones. We need to find compounds that satisfy the conditions for a positive iodoform test but a negative Tollen's test.

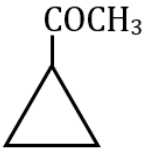
Let's examine each compound:

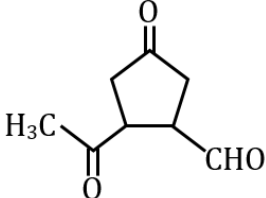
A. (a) CH_3CHO : Positive for both Tollen's and iodoform.

B. (b) Propanone: Positive for iodoform, negative for Tollen's.

C. (c) 2-Butanol: Positive for iodoform, negative for Tollen's.

D. (d)  : Positive for both Tollen's and iodoform.

E. (e)  : Positive for iodoform, negative for Tollen's.

F. (f)  : Positive for both iodoform and Tollen's.

G. (g) $\text{CH}_3\text{CH}_2\text{OH}$: Positive for iodoform, negative for Tollen's.

H. (h) CH_3COOH : Negative for both.

I. (i) Phenyl methyl ketone: Positive for iodoform, negative for Tollen's.

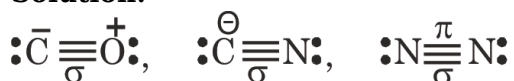
The compounds that give a positive iodoform test but a negative Tollen's test are (b), (c), (e), (g), and (i).

Final Answer: 5.00

24) **Question Explanation:** Calculate $(x \cdot y + z)$ for the given species.

Concept: Bonding in Isoelectronic Species

Solution:



$$x = 3 \quad y = 6 \quad \ell p = 6 = z \quad \Rightarrow (x \cdot y + z) \\ \Rightarrow (3 \times 6 + 6) = 24$$

The value of $(x \cdot y + z) = (3 \cdot 6 + 6) = 24$

Final Answer:(24.00)

25) **Question Explanation:** Ratio of $\psi^2_{r=a_0}$ to $\psi^2_{r=0}$

Concept : This question is based on Schrodinger wave equation.

Solution:

Probability density = ψ^2

$$\psi^2_{at(r=a_0)} = \left[\frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-\frac{a_0}{a_0}} \right]^2$$

$$\psi^2_{at(r=0)} = \left[\frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-\frac{0}{a_0}} \right]^2$$

$$\frac{\psi^2_{at(r=a_0)}}{\psi^2_{at(r=0)}} = e^{-2} = 0.14$$

Final Answer = 0.14

26) **Explain the question** Calculate boiling point elevation.

Concept This question is based on Van't Hoff Factor & Ebullioscopy.

Solution

$$\Delta T_b = i \times K_b \times m,$$

Copper(II) chloride (CuCl_2) is a strong electrolyte that completely dissociates in water into three ions ($\text{CuCl}_2 \rightarrow \text{Cu}^{2+} + 2\text{Cl}^-$), making the van't Hoff factor, $i = 3$.

CuCl_2 moles = $13.44 \text{ g} / 134.4 \text{ g/mol} = 0.1 \text{ mol}$.

molality of the solution: $m = \text{moles of solute} / \text{mass of solvent (kg)} = 0.1 \text{ mol} / 1 \text{ kg} = 0.1 \text{ m}$.

Finally, substitute the values into the formula: $\Delta T_b = 3 \times (0.52 \text{ K}\cdot\text{kg/mol}) \times (0.1 \text{ mol/kg}) = 0.156 \text{ K} \approx 0.16 \text{ K}$

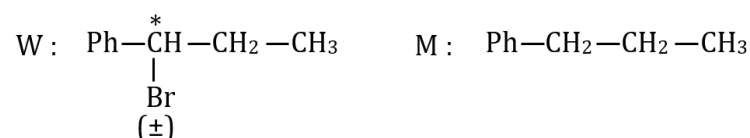
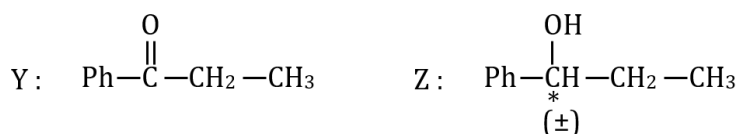
The elevation in boiling point is **0.16 K**

Final Answer (0.16)

27) **Explain Question:** Identify the product Z, W & M and reactant Y in the given sequence of reaction.

Concept: Apply concept of reduction, addition.

Solution:



Final Answer: (ACD)

28) **Explain Question:** Identify product P₁ & P₂ in the given sequence of reaction.

Concept: Dehydration is done by Conc. H₂SO₄

Solution:

P₁ : H₃C-CH₂-O-CH₂-CH₃ (diethyl ether)

P₂ : H₂C = CH₂ (ethene)

Final Answer: (BC)

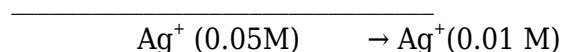
29) **Explain Question:** Identify the correct statement regarding to given cell.

Concept: Working principle of concentration cell.

Solution:

Cathode (x) Ag⁺ (0.05 M) + e⁻ → Ag_(s)

Anode (y) Ag → Ag⁺(g) (0.01 M) + e⁻



$$E_{\text{cell}} = E_{\text{Cell}}^0 - \frac{-0.0591}{n} \log \frac{0.01}{0.05}$$

$$E_{\text{cell}} = \frac{-0.0591}{1} \log \frac{0.01}{0.05}$$

E_{cell} increases when [cathode] increases and that [anode] decreases.

Half cell X is cathode, while Y is anode.

Final Answer: (A,D)

30) **Explain Question:** Select correct option according to given thermodynamic condition.

Concept: Apply concept of adiabatic reversible expansion.

Solution:

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} \quad T_2 = 300 \left(\frac{1}{8} \right)^{\frac{4}{3}-1} = 150 \text{ K}$$

$$w = \Delta U = nC_v \Delta T$$

$$= 1 \times \frac{6R}{2} (150 - 300)$$

$$= 3 \times 2 \times (-150) = -900 \text{ Cal}$$

Final Answer: (AD)

31) **Explain Question:** Identify pairs of compound which produce same gas.

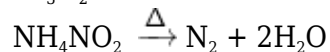
Concept: Heating effect on oxides and chlorides of nitrogen.

Solution:

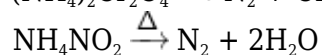
(B) both give N_2

(D) both give NH_3

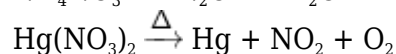
(A) $Ca_3N_2 + \text{dil HCl} \rightarrow CaCl_2 + NH_3$



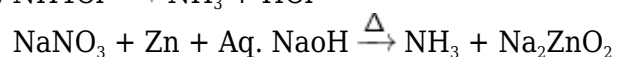
(B) $(NH_4)_2Cr_2O_4 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$



(C) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$



(D) $NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$



Final Answer: (BD)

32)

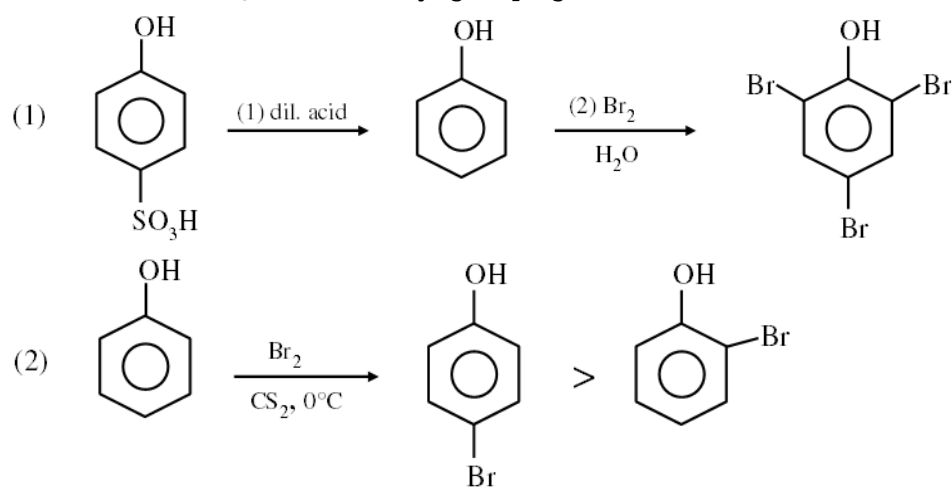
Explain Question: Question is asking about the reaction in which tribromo phenol is major product.

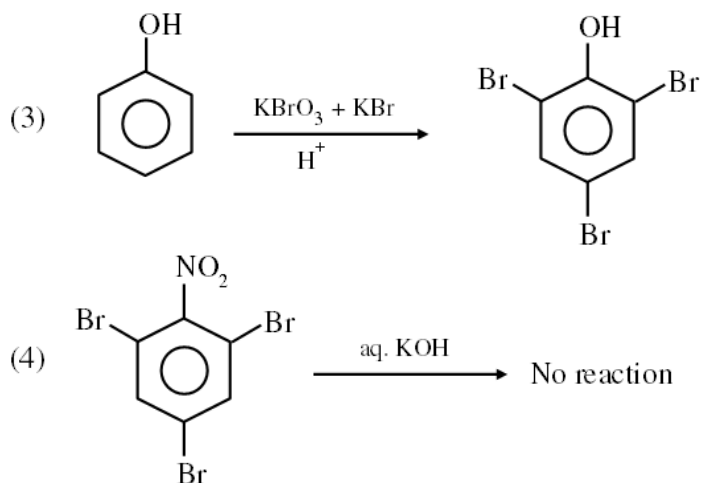
Concept :- This Question is based on electrophilic substitution reaction (ESR).

Solution :- Bromination of phenol in presence of water or with $(KBrO_3 + KBr + H^+)$ will form tribromophenol as major product.

IPSO substitution is also seen in bromination of phenol.

[i.e. $-COOH$, $-SO_3H$ and 3° alkyl groups give IPSO substitution].

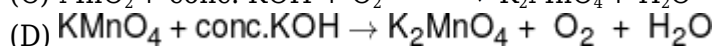
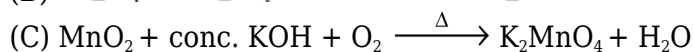
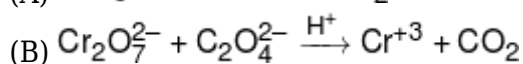
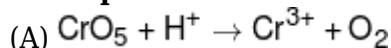




Final Answer :- (AC)

33) **Question Explanation:** Matching chemical reactions with their observations or types.

Concept: Reactions of d-Block Elements.



P→(1,2,3); Q→(1,3); R→(1,3); S→(1,2,3,).

The options are not provided, so a definitive match is not possible.

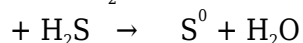
Final Answer: (C)

34)

Explaining the Question :- Match the oxidising agent of list-I with their characteristic product of list-II.

Concept:- Redox titration

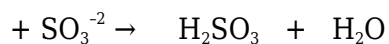
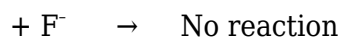
Solution :-



turbidity

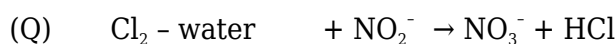


Red-brown

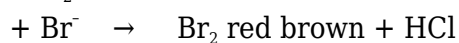
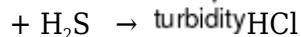


P → 2, 3, 5

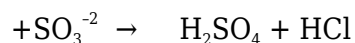
colorless colorless



S ↓



Q - 2, 3, 5



R → 1, 2, 3 S → 1, 2, 3 since both are only oxidising agent with coloured products.

Final Answer: (C)

35)

Explaining the Question :- Match the half life in List-I with order in List-II Half life of reaction.

Concept:- This question is based on

For first order $t_{1/2} \propto \frac{1}{a^0} = t_{1/2} = \text{Constant [I} \rightarrow \text{P]}$

for n^{th} order $t_{1/2} \propto \frac{1}{a^{n-1}}$
 $n = 0$ order $t_{1/2} \propto a$ [II $\rightarrow$ S, T]

$n = 2$ order $t_{1/2} \propto \frac{1}{a}$ [III $\rightarrow$ R]

$n = 3$ order $t_{1/2} \propto \frac{1}{a^2}$ [IV $\rightarrow$ Q]

Final Answer :- (C)

36)

A. Question Explanation: The question asks us to match compounds in List-I with properties in List-II, and then identify the incorrect matching. We need to understand the types of hydrogen bonding and how they affect properties like vapor pressure and apparent molecular mass.

B. Given Data:

List-I:

- A. (I) o-nitrophenol
- B. (II) Chloral hydrate
- C. (III) p-hydroxybenzaldehyde
- D. (IV) Acetic acid

List-II:

- A. (P) Intermolecular H-bonding
- B. (Q) Intramolecular H bonding
- C. (R) Higher vapour pressure than its o-/p- form, if they exist for the molecule
- D. (S) Chlorine forms H-bond in this compound
- E. (T) Molecular mass is found twice than ideal experimentally
- F. (U) Hydrogen forming H-bond is bonded with oxygen in this compound

C. Concept: Intramolecular H-bonding and Intermolecular H-bonding.

D. Solution:

- A. **Intramolecular H-bonding:** Hydrogen bonding within the same molecule. This often occurs when a hydrogen atom is bonded to a highly electronegative atom (like O, N, or F) and is also close to another electronegative atom in the same molecule. It can affect shape and properties like boiling point (generally lowers it).
- B. **Intermolecular H-bonding:** Hydrogen bonding between different molecules. This increases the attractive forces between molecules, leading to higher boiling points and sometimes

affecting apparent molecular mass (due to association).

- C. **Vapor Pressure:** Stronger intermolecular forces (like those from hydrogen bonding) lead to lower vapor pressure.
- D. **(I) o-nitrophenol:** Exhibits intramolecular H-bonding between the -OH and the nitro group's oxygen.
- E. **(II) Chloral hydrate:** Has intramolecular H-bonding. The presence of three highly electronegative chlorine atoms on the same carbon makes the hydrogen on the adjacent carbon very positive and it forms an intramolecular hydrogen bond with one of the oxygen atoms of the hydroxyl group.
- F. **(III) p-hydroxybenzaldehyde:** Exhibits intermolecular H-bonding.
- G. **(IV) Acetic acid:** Forms intermolecular H-bonds, leading to dimerization and an apparent doubling of molecular mass in some situations.

Now let's match and find the incorrect combination:

- A. I - Q (Correct - Intramolecular H-bonding in o-nitrophenol)
- B. II - Q (Correct - Intramolecular H-bonding in chloral hydrate)
- C. III - P (Correct - Intermolecular H-bonding in p-hydroxybenzaldehyde)
- D. IV - T (Correct - Dimerization of acetic acid leading to a higher apparent molecular weight)

The options given in the question are not the pairings themselves, but rather the row numbers. All the individual pairings are correct. The question asks for the incorrect combination. Since all the pairings are correct, there is no incorrect combination. However, if the question is asking which option contains a pairing that is NOT a property of the compound, then the answer is (R).

E. Final Answer: (C)

Since all the pairings are correct, there is no incorrect combination among the given options. If the intent of the question was to find a property that does not apply, then option 1 (R) would be the answer, as neither o-nitrophenol nor p-hydroxybenzaldehyde has a higher vapor pressure than their other isomers due to hydrogen bonding.

PART 3 : MATHEMATICS

37) **Explain Question :** We have to solve the integration and find m

Concept : Properties of DI

Solution :

$$\begin{aligned}
 & \int_0^1 \frac{x}{1-x^4} (\ln x)^2 dx \\
 &= \int_0^1 (x + x^5 + x^9 + \dots \infty) (\ln x)^2 dx \\
 &= \int_0^1 \sum_{k=0}^{\infty} x^{4k+1} (\ln x)^2 dx \\
 &= \sum_{k=0}^{\infty} \int_0^1 x^{4k+1} (\ln x)^2 dx
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \sum_{k=0}^n \left[(\ln x)^2 \cdot \frac{x^{4K+2}}{4K+2} \Big|_0^1 - \int_0^1 \frac{2 \ln x}{x} \cdot \frac{x^{4K+2}}{(4K+2)} dx \right] \\
&= - \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_0^1 \frac{\ln x \cdot x^{4K+1}}{(2K+1)} dx \\
&= - \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_0^1 \frac{x^{4K+1} \cdot \ln x}{(2K+1)} dx \\
&= - \left(\lim_{n \rightarrow \infty} \sum_{k=0}^n \left[\ln x \cdot \frac{x^{4K+2}}{2(2K+1)^2} \Big|_0^1 - \int_0^1 \frac{1}{x} \cdot \frac{x^{4K+2}}{2(2K+1)^2} dx \right] \right) \\
&= \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_0^1 \frac{x^{4K+1}}{2(2K+1)^2} dx \\
&= \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{x^{4K+2}}{4(2K+1)^3} \Big|_0^1 = \frac{S_3}{4} = 0.25 S_3
\end{aligned}$$

Final Answer : (0.25)

38) **Explain Question :** We have to solve the integral and find λ

Concept : Integration by substitution

Solution :

Let $A = (x^2 + 1^2)(x^2 + 2^2) \dots (x^2 + 100^2)$

$\square nA = \square n(x^2 + 1^2) + \dots + \square n(x^2 + 100^2)$

differentiate w.r.t. x $\frac{1}{A} \frac{dA}{dx} = \frac{2x}{x^2 + 1^2} + \dots + \frac{2x}{x^2 + 100^2}$

$\square$ L.H.S. of given integral becomes

$$\begin{aligned}
&\int_0^1 \frac{1}{2A} \cdot \frac{dA}{dx} \cdot A dx = \frac{1}{2} \left[(x^2 + 1^2) (x^2 + 2^2) \dots (x^2 + 100^2) \right]_0^1 \\
&= \frac{1}{2} \left[\prod_{k=1}^{100} (1^2 + k^2) - (100)^2 \right] \\
&\therefore \lambda = 2 \text{ \& } \frac{1}{\lambda} = 0.5
\end{aligned}$$

Final Answer : (0.50)

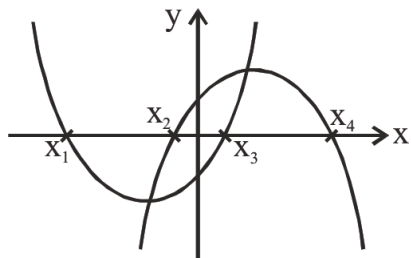
39) **Explain Question :** We have to find m, p, n using x_1, x_2, x_3 and x_4 .

Concept : Graph of quadratic expression

Solution :

Graph of both the quadratic functions are symmetric about the point $(2, 0)$. Shift both the graph horizontally so that they become symmetric about point $(0, 0)$ as the difference of roots remains invariant under such transformation.

Let translated versions of $f(x)$ and $g(x)$ are $u(x)$ and $v(x)$



$$\square x_3 = -x_2 = 3 \text{ and } x_4 = -x_1$$

$$\square u(x) = -v(-x).$$

$$\square u(x) = a(x-3)(x-x_1) \dots\dots(1)$$

$$\text{and } v(x) = a(x+3)(x+x_1) \dots\dots(2)$$

$$\text{put } x = \frac{3+x_1}{2} \text{ in (1)}$$

$$\square \text{ vertex of } u(x) \text{ lies on } v(x)$$

$$\square (2) = -\frac{a}{4}(x_1-3)^2 = a\left(\frac{3+x_1}{2}+3\right)\left(\frac{3+x_1}{2}+x_1\right)$$

$$\Rightarrow (x_1-3)^2 = (x_1+9)(3x_1+3)$$

$$\Rightarrow x_1^2 + 18x_1 + 9 = 0$$

$$\Rightarrow x_1 = -9 \pm 6\sqrt{2} \quad \square \quad x_1 < -3$$

$$\Rightarrow x_1 = -9 - 6\sqrt{2}$$

$$\square x_4 - x_1 = -x_1 - x_1 = -2x_1$$

$$= 18 + 12\sqrt{2} \Rightarrow \frac{\text{mp}}{n} = \frac{18 \times 2}{12} = 3$$

Final Answer : (3.00)

40) Explain Question : We have to find the value of the required expression

Concept : Adjoint of a matrix

Solution :

$$\text{Let } A = \begin{bmatrix} \sin \alpha & \cos \alpha & 0 \\ \cos \alpha & 0 & \sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now, } |A| = -[\sin^3 \alpha + \cos^3 \alpha] \quad \begin{matrix} 1 \\ -1 \end{matrix} \quad \left(\because \alpha = \frac{n\pi}{2} \right)$$

$$\Rightarrow |A| \neq 0 \Rightarrow X = A^{-1}B$$

$$\Rightarrow X = \frac{\text{adj } A \cdot B}{|A|}$$

$$\Rightarrow \frac{1}{|A|} \begin{bmatrix} -\sin^2 \alpha & - & - \\ -\cos^2 \alpha & - & - \\ \sin \alpha \cos \alpha & - & - \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow x = -\frac{\sin^2 \alpha}{|A|}, y = -\frac{\cos^2 \alpha}{|A|}, z = \frac{\sin \alpha \cos \alpha}{|A|}$$

$$\text{Now, } x - y = \frac{\cos^2 \alpha - \sin^2 \alpha}{|A|} = \frac{\cos 2\alpha}{|A|}$$

$$\begin{aligned} \text{and } 2z &= \frac{\sin 2\alpha}{|A|} \\ \Rightarrow (x-y)^2 + 4z^2 &= \frac{\cos^2 2\alpha + \sin^2 2\alpha}{|A|^2} \\ \Rightarrow \frac{1}{|A|^2} &= 1 \quad (\because |A| = 1 \text{ or } -1) \end{aligned}$$

Final Answer : (1.00)

41) **Explain Question :** We have to solve the integration to find λ .

Concept : Properties of DI.

Solution :

$$\begin{aligned} &\int_0^1 \frac{\log_e x}{1-x^2} dx \\ &= \left(\log_e x \cdot \frac{1}{2} \log_e \left| \frac{1+x}{1-x} \right| \right)_0^1 - \frac{1}{2} \int_0^1 \log_e \left| \frac{1+x}{1-x} \right| \cdot \frac{1}{x} dx \\ &= \frac{1}{2} \left(\frac{\log_e \left(\frac{1+x}{1-x} \right)}{\frac{1}{\log_e x}} \right)_0^1 - \frac{1}{2} \int_0^1 \log_e \left(\frac{1+x}{1-x} \right) \frac{dx}{x} \\ &= 0 - \frac{1}{2} \int_0^1 \frac{\log_e(1+x) - \log_e(1-x)}{x} dx \\ &= - \int_0^1 \left(1 + \frac{x^2}{3} + \frac{x^4}{5} + \dots \right) dx \\ &= - \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right) = \frac{-\pi^2}{8} \\ &\left(\frac{1}{1^2} + \frac{1}{3^2} + \dots \infty \right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty \right) = \frac{\pi^2}{6} \\ \Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \dots \infty &= \frac{\pi^2}{8} \end{aligned}$$

Final Answer : (8.00)

42) **Explain Question :** We have to find m and n

Concept : One-one function

Solution :

$$4^{f(x)} + 2^{-x} = 8$$

$$\Rightarrow f(x) = \log_4(8 - 2^{-x})$$

$$\text{domain } 8 - 2^{-x} > 0 \Rightarrow 2^{-x} < 2^3$$

$$\Rightarrow x \in (-3, \infty) \text{ So, } m = -2$$

$$\text{For range } f(x)_{\max} = \log_4(8) = \frac{3}{2}$$

$$\square \text{ range } \left(-\infty, \frac{3}{2} \right) \text{ So, } n = 1$$

$$\left| \frac{n}{m} \right| = \left| -\frac{1}{2} \right| = 0.50$$

$$4^{f(x)} + 2^{-x} = 8$$

$$\Rightarrow 4^{f(x)} \cdot \ln 4 \cdot f'(x) - 2^{-x} \ln 2 = 0$$

$$\Rightarrow f'(x) = \frac{2^{-x} \ln 2}{\ln 4 \cdot 4^{f(x)}} = \frac{2^{-x}}{2 \cdot 4^{f(x)}}$$

$$\Rightarrow f'(x) = \frac{2^{-1-x}}{4^{f(x)}} = \frac{1}{2^{1+x+2f(x)}}$$

$$\Rightarrow f'(x) > 0 \quad \forall x \in D_f$$

So, $f(x)$ is one-one function

Final Answer : (0.50)

43) **Explain Question :** We have to solve the given limit

Concept : Evaluating limits by substitution

Solution :

$$x = \tan \theta$$

$$\begin{aligned} L &= \lim_{\theta \rightarrow 0} \frac{1}{\ln(\tan \theta + \sec \theta)} - \frac{1}{\ln(1 + \tan \theta)} \\ \Rightarrow L &= \lim_{\theta \rightarrow 0} \frac{\ln(1 + \tan \theta) - \ln(\tan \theta + \sec \theta)}{\ln(\tan \theta + \sec \theta) \cdot \ln(1 + \tan \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{\ln(\sin \theta + \cos \theta) - \ln(1 + \sin \theta)}{(\tan \theta) (\ln(\sec \theta + \tan \theta))} \\ &= \lim_{\theta \rightarrow 0} \frac{\frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta} - \frac{\cos \theta}{1 + \sin \theta}}{(\sec^2 \theta) (\ln(\sec \theta + \tan \theta)) + (\tan \theta) \sec \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{\cos \theta - \sin \theta - 1}{\ln(1 + \sin \theta) - \ln \cos \theta + \sin \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{-\sin \theta - \cos \theta}{\frac{\cos \theta}{1 + \sin \theta} + \frac{\sin \theta}{\cos \theta} + \cos \theta} = -\frac{1}{1+1} = -\frac{1}{2} \end{aligned}$$

Final Answer : (-.50)

44) **Explain Question :** We have to evaluate the limit

Concept : Definite integral as limit of sum

Solution :

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{\pi}{2n} \left(1 + \cos \frac{\pi}{2n} + \cos \frac{2\pi}{2n} + \dots + \cos \frac{(n-1)\pi}{2n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\pi}{2n} \left(\sum_{r=1}^n \cos \frac{(r-1)\pi}{2n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\pi}{2n} \left(\sum_{r=1}^n \cos \left(\frac{\pi r}{2n} - \frac{\pi}{2n} \right) \right) \\ &= \lim_{n \rightarrow \infty} \frac{\pi}{2n} \left[\sum_{r=1}^n \left(\cos \frac{\pi r}{2n} \cos \frac{\pi}{2n} + \sin \frac{\pi r}{2n} \sin \frac{\pi}{2n} \right) \right] \\ &= \lim_{n \rightarrow \infty} \frac{\pi}{2} \cos \frac{\pi}{2n} \left(\sum_{r=1}^n \frac{1}{n} \cos \frac{\pi r}{2n} \right) + \lim_{n \rightarrow \infty} \frac{\pi}{2} \sin \frac{\pi}{2n} \left(\sum_{r=1}^n \frac{1}{n} \sin \frac{\pi r}{2n} \right) \quad \dots(1) \end{aligned}$$

$$\text{as } n \rightarrow \infty, \quad \cos \frac{\pi}{2n} \rightarrow 1$$

$$\text{as } n \rightarrow \infty, \quad \sin \frac{\pi}{2n} \rightarrow 0$$

Using Integral as limit of a sum-

replace :- $\frac{1}{n}$ by dx .
 $\frac{r}{n}$ by x .

$\lim_{n \rightarrow \infty} \sum$ by $\int$ sign.

Lower limit $\Rightarrow r = 1, n \rightarrow \infty$

$$\frac{r}{n} = \frac{1}{n} = 0$$

upper limit $\Rightarrow r = n, n \rightarrow \infty$

$$\frac{r}{n} = \frac{n}{n} = 1$$

From equation (1)

$$= \frac{\pi}{2} \int_0^1 \cos \frac{\pi x}{2} \cdot dx - 0$$

$$= \frac{\pi}{2} \int_0^1 \left[\frac{\sin \pi x / 2}{\pi / 2} \right]_0^1$$

$$= \left[\sin \frac{\pi x}{2} \right]_0^1 = 1 - 0 = 1$$

Final Answer : (1.00)

45) **Explain Question :** We have to solve the following equations to find the number of solutions

Concept : Basic trigonometric identities

Solution :

$$\log_2 \left(\frac{1 - \sin x}{\cos x} \right) + \log_4 \left(\frac{1 + \sin^2 x + 2 \sin x}{\cos^2 x} \right)$$

$$= \log_2 \left(\frac{1 - \sin x}{\cos x} \right) + \log_{2^2} \left(\frac{1 + \sin x}{\cos x} \right)^2$$

$$= \log_2 \left(\frac{1 - \sin x}{\cos x} \right) + \log_2 \left(\frac{1 + \sin x}{\cos x} \right)$$

$$= \log_2 \frac{1 - \sin^2 x}{\cos^2 x} = \log_2 1 = 0$$

□ Given equation reduces to.

$$2^{\log_2 \cos x} = (\sin x + \operatorname{cosec} x)^{\log_2 \operatorname{cosec} x}$$

$$\Rightarrow \cos x = (\sin x + \operatorname{cosec} x)^{\log_2 \operatorname{cosec} x}$$

$$\Rightarrow \cos x = 1 \text{ or } \sin x + \operatorname{cosec} x = 2$$

both gives $\sin x = 1$ which does not satisfy the parent equation

$\Rightarrow$ Number of values of $x = 0$

Now check each option.

Final Answer : (A,B,D)

46) **Explain Question :** We have to solve for $f(x)$

Concept : Composite function

Solution :

Let $k = f(x) + \frac{1}{x}$. Then $k > 0$ (given)...(i)

$$\text{so } k \text{ will satisfy } f(x) \left[f \left(f(x) + \frac{1}{x} \right) \right] = 1$$

$$\Rightarrow f(k) \left[f \left(f(k) + \frac{1}{k} \right) \right] = 1 \quad \dots(ii)$$

$$\text{Also } f(x).f(k) = 1 \quad \dots(iii)$$

by comparing (ii) and (iii) we say

$$f(x) = f \left(f(k) + \frac{1}{k} \right)$$

$$f(x) = f \left(\frac{1}{f(x)} + \frac{1}{f(x) + \frac{1}{x}} \right) \quad (\text{by i and iii})$$

as f is injective function

$$x = \frac{1}{f(x)} + \frac{x}{xf(x) + 1}$$

$$x^2(f(x))^2 - xf(x) - 1 = 0$$

$$f(x) = \frac{x \pm \sqrt{x^2 + 4x^2}}{2x^2} = \frac{1 \pm \sqrt{5}}{2x}$$

$$\text{only } f(x) = \frac{1 - \sqrt{5}}{2x} \text{ satisfy all three given conditions.}$$

Final Answer : (B,D)

47) **Explain Question :** We have to solve for A and B

Concept : Special sequences

Solution :

$$S = \sum_{n=1}^8 \sqrt{3 + \sqrt{n}}, \quad T = \sum_{n=1}^8 \sqrt{3 - \sqrt{n}}$$

Let

Note that $\sqrt{a + b + 2\sqrt{ab}} = \sqrt{a} + \sqrt{b}$ for positive real numbers a and b . In particular, for each fixed positive integer n with $1 \leq n \leq 8$ choose $a > b > 0$ satisfying $a + b = 6$, $ab = n$ then a, b are roots of the equations $x^2 - 6x + n = 0$ i.e.

$$a = 3 + \sqrt{9 - n}, \quad b = 3 - \sqrt{9 - n}$$

$$\text{so } \sqrt{2}S = \sum_{n=1}^8 \sqrt{6 + 2\sqrt{n}}$$

$$= \sum_{n=1}^8 \left(\sqrt{3 + \sqrt{9 - n}} + \sqrt{3 - \sqrt{9 - n}} \right)$$

$$= \sum_{n=1}^8 \left(\sqrt{3 + \sqrt{n}} + \sqrt{3 - \sqrt{n}} \right) = S + T$$

$$\text{Thus, } (\sqrt{2} - 1)S = T \Rightarrow \frac{S}{T} = \frac{1}{\sqrt{2} - 1} = 1 + \sqrt{2}$$

Final Answer : (B,C,D)

48) **Explain Question:** We have to solve for α and β .

Concept: Euler form.

Solution:

$$z = e^{i \frac{2\pi}{2n+1}}$$

$\square 1, z, z^2, z^3, \dots, z^{2n}$ denotes $2^{n+1}, (2n + 1)^{\text{th}}$ roots of unity.

Sum of roots of given equation

$$\alpha + \beta = z + z^2 + z^3 + \dots + z^{2n-1} + z^{2n} = -1$$

Product of roots

$$\begin{aligned}\alpha\beta &= (z^2 + z^4 + \dots + z^{2n})(z + z^3 + \dots + z^{2n-1}) \\ &= z^2(1 + z^2 + z^4 + \dots + z^{2n-2}) \cdot z(1 + z^2 + z^4 + \dots + z^{2n-2}) \\ &= z^3(1 + z^2 + z^4 + \dots + z^{2n-2})^2 \\ &= z^3 \left(\frac{1 - ((z^2)^n - 1)}{z^2 - 1} \right)^2 \\ \Rightarrow \alpha\beta &= z^3 \left(\frac{z^{2n} - 1}{z^2 - 1} \right)^2 \cdot \frac{1}{(z + 1)^2} \dots(1)\end{aligned}$$

$$\text{Now } 1 + z + z^2 + \dots + z^{2n} = \frac{z^{2n+1} - 1}{z - 1} = 0$$

$$\Rightarrow \frac{z^{2n+1} - z + z - 1}{z - 1} = 0$$

$$\Rightarrow \frac{z(z^{2n} - 1)}{z - 1} = -1$$

$$\Rightarrow \frac{z^{2n} - 1}{z - 1} = \frac{-1}{z} \dots(2)$$

put (2) in (1)

$$\alpha\beta = z^3 \left(\frac{-1}{z} \right)^2 \cdot \frac{1}{(z + 1)^2}$$

$$\Rightarrow \alpha\beta = \frac{1}{(z + 1)^2}$$

$$\alpha\beta = \frac{1}{z + \frac{1}{z} + 2}$$

$$\Rightarrow \alpha\beta = \frac{1}{2\cos\theta + 2}, \text{ where } \theta = \frac{2\pi}{2n+1}$$

$$\Rightarrow \alpha\beta = \frac{1}{2(2\cos^2\frac{\theta}{2})} = \frac{\sec^2\frac{\theta}{2}}{4}$$

$$\Rightarrow \alpha\beta = \frac{1}{4}\sec^2\left(\frac{\pi}{2n+1}\right)$$

□ Equation can be written as

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (-1)x + \frac{1}{4}\sec^2\left(\frac{\pi}{2n+1}\right) = 0$$

Final Answer : (A,C)

49) **Explain Question:** We have to solve for f(x).

Concept: Linear DE and finding a function.

Solution:

$$\int_1^x f(x) dx = \frac{x^2 + 1}{2x} \left(f(x) - x + \frac{2x}{x^2 + 1} \right) - \log_e 2$$

$$\int_1^x f(x) dx = \frac{x^2+1}{2x} f(x) - \left(\frac{x^2+1}{2} \right) + 1 - \log_e 2$$

$$f(x) = \frac{1}{2} \left(1 - \frac{1}{x^2} \right) f(x) + \left(\frac{x^2+1}{2x} \right) f'(x) - x$$

$$\frac{f(x)(x^2+1)}{2x^2} = \left(\frac{x^2+1}{2x} \right) f'(x) - x$$

$$f(x) = x f'(x) - \frac{2x^3}{x^2+1}$$

$$(A) \quad g(x) = \frac{x f'(x) - f(x)}{x} = \frac{2x^2}{x^2+1} \in [0, 2)$$

$$x \frac{dy}{dx} - y = \frac{2x^3}{x^2+1}$$

$$\frac{dy}{dx} - \frac{y}{x} = \frac{2x^2}{x^2+1} \Rightarrow \frac{y}{x} = \log(x^2+1) + c$$

$$f(1) = \log_e 2 \Rightarrow c = 0$$

$$f(x) = x \log_e(x^2+1)$$

(B)

$$(C,D) \quad f(x) = x \log(x^2+1)$$

$$\Rightarrow f'(x) = \log(x^2+1) + \frac{2x^2}{x^2+1}$$

$$f'(x) - f(x) = (1-x) \log(x^2+1) + \frac{2x^2}{x^2+1}$$

$$f'(x) - f(x) < 0$$

for atleast one $x = \alpha, \alpha \in (1, \infty)$

Final Answer : (A,B,D)

50) **Explain Question:** We have to find $f(x)$.

Concept: Estimation of DI.

Solution:

$$\int_0^1 x^n f(x) dx = 0 \quad \forall n \in \mathbb{N}_0$$

Since 0

$$\int_0^1 f(x) P(x) dx = 0; \quad P(x)$$

is any polynomial in 'x'

$$I = \int_0^1 f^2(x) dx = \int_0^1 f^2(x) dx - \int_0^1 f(x) P(x) dx$$

Consider

$$= \int_0^1 f(x) (f(x) - P(x)) dx$$

$$= \int_0^1 f(x) dx = 0, \text{ for } n = 0$$

Further,

$\Rightarrow f(x)$ must be bounded

$\Rightarrow f(x) - P(x)$ must be bounded as $P(x)$ is a polynomial

Hence $|f(x) - P(x)| \leq m$ for some $m \in \mathbb{R}_0^+$ and $\forall x \in [0, 1]$

$$I = \int_0^1 f(x)(f(x) - P(x))dx \leq \int_0^1 |f(x) - P(x)| dx \leq \int_0^1 m f(x) dx = m \int_0^1 f(x) dx = 0$$

So,

$$\Rightarrow I \leq 0 \Rightarrow \int_0^1 f^2(x) dx \leq 0$$

$$\Rightarrow \int_0^1 f^2(x) dx = 0 \Rightarrow f(x) = 0 \forall x \in [0, 1]$$

Final Answer : (A,B)

51) **Explain Question :** We have to solve the summations

Concept : Special sequences

Solution :

$$(P) \quad \tan \frac{x}{2} = \cot \frac{x}{2} - 2 \cot x$$

We know, $\tan \frac{x}{2} = \cot \frac{x}{2} - 2 \cot x$

$$\frac{1}{2} \tan \frac{\pi}{2^3} = \frac{1}{2} \left[\cot \frac{\pi}{2^3} - 2 \cot \frac{\pi}{2^2} \right]$$

$$\frac{1}{2^2} \tan \frac{\pi}{2^4} = \frac{1}{2^2} \left[\cot \frac{\pi}{2^4} - 2 \cot \frac{\pi}{2^3} \right]$$

$$\frac{1}{2^3} \tan \frac{\pi}{2^5} = \frac{1}{2^3} \left[\cot \frac{\pi}{2^5} - 2 \cot \frac{\pi}{2^4} \right]$$

$$\frac{1}{2^n} \tan \frac{\pi}{2^{n+2}} = \frac{1}{2^n} \left[\cot \frac{\pi}{2^{n+2}} - 2 \cot \frac{\pi}{2^{n+1}} \right]$$

$$\text{Sum} = \lim_{n \rightarrow \infty} \frac{1}{2^n} \left[\cot \frac{\pi}{2^{n+2}} \right] - \cot \frac{\pi}{4}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2^n \left(\frac{\tan \frac{\pi}{2^{n+2}}}{\frac{\pi}{2^{n+2}}} \right) \left(\frac{\pi}{2^{n+2}} \right)} - 1$$

$$= \frac{4}{\pi} - 1$$

$$(Q) \quad = \sum_{k=1}^{\infty} \tan^{-1} \left(\frac{4}{4k^2 + 3} \right)$$

$$\tan^{-1} \left(\frac{4}{4 + (2k+1)(2k-1)} \right) = \tan^{-1} \left(\frac{1}{1 + \frac{(2k+1)(2k-1)}{2}} \right)$$

$$= \tan^{-1} \frac{2k+2}{2} - \tan^{-1} \frac{2k-1}{2}$$

$$T_1 = \tan^{-1} \frac{3}{2} - \tan^{-1} \frac{1}{2}$$

$$T_2 = \tan^{-1} \frac{5}{2} - \tan^{-1} \frac{3}{2}$$

⋮
⋮
⋮

$$T_n = \tan^{-1} \frac{2n+1}{2} - \tan^{-1} \frac{2n-1}{2}$$

$$S_n = \tan^{-1} \frac{2n+1}{2} - \tan^{-1} \frac{1}{2}$$

$$S_\infty = \frac{\pi}{2} - \tan^{-1} \frac{1}{2} = \cot^{-1} \frac{1}{2} \\ = \tan^{-1} 2$$

$$(R) \sum_{k=1}^{\infty} \frac{1}{(2k-1)2k(2k+1)} \\ = \frac{1}{2} \sum_{k=1}^{\infty} \left(\frac{1}{2k-1} + \frac{1}{2k+1} - \frac{2}{2k} \right) \text{ which is the expansion of } \ln(1+x) - \frac{1}{2} \text{ for } x=1$$

$$(S) \sum_{k=1}^{\infty} \frac{1}{(4k-1)(4k+1)} \\ = \frac{1}{2} \sum_{k=1}^{\infty} \left(\frac{1}{4k-1} - \frac{1}{4k+1} \right) \\ = \frac{1}{2} \left\{ \frac{1}{3} - \frac{1}{5} + \frac{1}{7} - \frac{1}{9} + \dots \right\} \dots \\ = \frac{1}{2} (1 - \tan^{-1} 1) \\ = \frac{1}{2} \left(1 - \frac{\pi}{4} \right) \\ = \frac{1}{2} - \frac{\pi}{8}$$

Final Answer : (A)

52) **Explain Question :** We have to solve the summation and limit

Concept : Sandwich theorem and methods to find limit

Solution :

$$6 = 5e - 19 \Rightarrow e = 5$$

$$4e - 19 = 1 - 4d \Rightarrow d = 0$$

$$1 - 3d = 3c - 5 \Rightarrow c = 1$$

$$1 - b = 2a - 1 \Rightarrow a = \frac{1}{2}$$

(P) For every $x \geq 0$, we have $x - x^3 \leq \sin x \leq x$

$$\Rightarrow \frac{i}{n^2} - \frac{i^3}{n^6} \leq \sin \left(\frac{i}{n^2} \right) \leq \frac{i}{n^2}$$

$$\Rightarrow \frac{n(n+1)}{2n^2} - \frac{n^2(n+1)^2}{4n^6} \leq \sum_{n=1}^n \sin \left(\frac{i}{n^2} \right) \leq \frac{n(n+1)}{2}$$

$$\left\{ \begin{array}{l} \because \lim_{n \rightarrow \infty} \left(\frac{n(n+1)}{2n^2} - \frac{n^2(n+1)^2}{4n^6} \right) = \frac{1}{2} \\ \because \lim_{n \rightarrow \infty} \frac{n(n+1)}{2} = \frac{1}{2} \end{array} \right\}$$

$$\lim_{n \rightarrow \infty} \sum_{n=1}^n \sin \left(\frac{i}{n^2} \right) = \frac{1}{2}$$

(Q) We know that $[x] + \left[x + \frac{1}{2}\right] = [2x]$

$$\therefore \left[\frac{n}{2^k} + \frac{1}{2}\right] = \left[\frac{n}{2^{k-1}}\right] - \left[\frac{n}{2^k}\right]$$

$$\Rightarrow \sum_{k=1}^n \left[\frac{n}{2^k} + \frac{1}{2}\right] = [n] - \left[\frac{n}{2^k}\right] = n$$

$$\sum \left(\sqrt{1 + \frac{k}{n^2}} - 1\right) = \sum \frac{\frac{k}{n^2}}{\sqrt{1 + \frac{k}{n^2}} + 1} = S_n(\text{Say})$$

(R)

$$\sum \frac{\frac{k}{n^2}}{\sqrt{1 + \frac{k}{n^2}} + 1} \leq S_n \leq \sum \frac{\frac{k}{n^2}}{\sqrt{1 + \frac{1}{n^2}} + 1}$$

$$\Rightarrow \frac{n(n+1)}{2n^2 \left[1 + \sqrt{1 + \frac{1}{n}}\right]} \leq S_n \leq \frac{n(n+1)}{2n^2 \left(1 + \sqrt{1 + \frac{1}{n^2}}\right)}$$

$$\left\{ \begin{array}{l} \therefore \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2 \left[1 + \sqrt{1 + \frac{1}{n}}\right]} = \frac{1}{2} \\ \therefore \lim_{n \rightarrow \infty} S_n = \frac{1}{2} \\ \therefore \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2 \left(1 + \sqrt{1 + \frac{1}{n^2}}\right)} = \frac{1}{2} \end{array} \right\}$$

(S) $\sum \frac{r}{n^2 + n + 2n} < \sum \frac{r}{n^2 + n + 2r} < \sum \frac{r}{n^2 + n + 2}$

$$\Rightarrow \frac{n(n+1)}{2(n^2 + 3n)} < \sum \frac{r}{n^2 + n + 2r} < \frac{n(n+1)}{2(n^2 + n + 2)}$$

$$\left\{ \begin{array}{l} \therefore \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2 + 3n)} = \frac{1}{2} \\ \therefore \lim_{n \rightarrow \infty} \sum \frac{r}{n^2 + n + 2r} = \frac{1}{2} \\ \therefore \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2 + n + 2)} = \frac{1}{2} \end{array} \right\}$$

Final Answer : (B)

53) **Explain Question :** We have to find different properties of $h(x)$

Concept : Types and differentiability of functions

Solution :

$$f(x) = \begin{cases} x|x| \sin \frac{1}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases} \quad g(x) = \begin{cases} 1 - 2x & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$

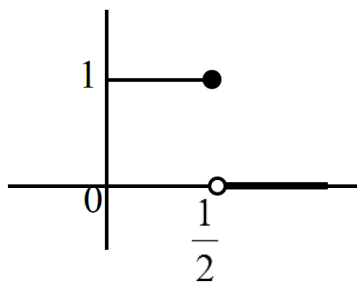
$$g\left(\frac{1}{2} - x\right) = \begin{cases} 2x & ; 0 \leq \frac{1}{2} - x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases} = \begin{cases} 2x & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$

$$g(x) + g\left(\frac{1}{2} - x\right) = \begin{cases} 1 & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$

(P) Now $a = 0, b = 1, c = 0, d = 0$

$$h(x) = g(x) + g\left(\frac{1}{2} - x\right) = \begin{cases} 1 & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$

□



Hence Range of $h(x)$ is $\{0, 1\}$

(Q) $a = 1, b = 0, c = 0, d = 0$

$$f(x) = \begin{cases} x|x| \sin \frac{1}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$h(x) =$

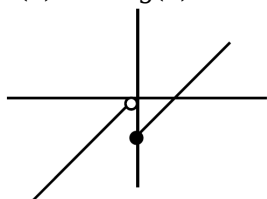
$$\text{RHD} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x} - 0}{x} = 0$$

$$\text{LHD} = \lim_{x \rightarrow 0} \frac{-x^2 \sin \frac{1}{x} - 0}{x} = 0$$

Hence $h(x)$ is differentiable on $\mathbb{R}$

(R) $a = 0, b = 0, c = 1, d = 0$

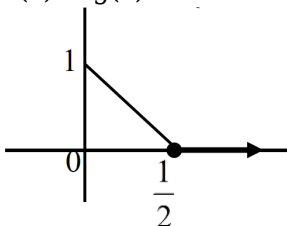
$$h(x) = x - g(x) = \begin{cases} 3x - 1 & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$



$\square$ $h(x)$ is ONTO

(S) $a = 0, b = 0, c = 0, d = 1$

$$h(x) = g(x) = \begin{cases} 1 - 2x & ; 0 \leq x \leq \frac{1}{2} \\ 0 & ; \text{otherwise} \end{cases}$$



Range of $h(x)$ is $[0, 1]$

Final Answer : (C)

54) **Explain Question:**

(P) We have to find I.

(Q) We have to find the required probability.

(R) We have to find number of solutions.

(S) Check for continuity.

Concept:

(P) Properties of DI.

(Q) Basics of probability.

(R) Graphs of functions.

(S) Continuity of function.

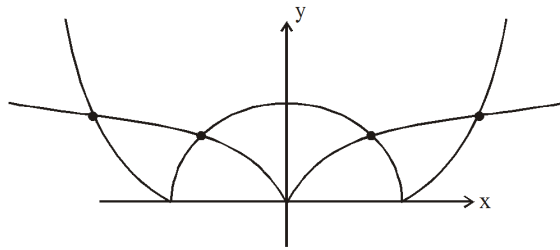
Solution:

(P) $I = \int_0^{\infty} \lfloor e^{1-x} \rfloor dx. 0 < x < \infty \Rightarrow -\infty < 1-x < 1 \Rightarrow 0 < e^{1-x} < e$

Now $0 < e^{1-x} < 1 \Leftrightarrow -\infty < 1-x < 0 \Rightarrow 1 < x < \infty$. Similarly $1 < e^{1-x} < 2$ for $1 - \ln 2 < x < 1$ and $2 < e^{1-x} < e$ for $0 < x < 1 - \ln 2$.

So $I = \int_0^{1-\ln 2} 2 dx + \int_{1-\ln 2}^1 1 dx + \int_1^{\infty} 0 dx$

(Q) We are unable to form groups that do not satisfy the given condition. So we try to prove that the given condition will always be satisfied. If possible, let there be three groups such that the product in each group ≤ 71 . $\Rightarrow$ product of all numbers $\leq 71^3$ but $9!$ turns out to be greater than 71^3 . This is a contradiction. Hence required probability is 1



(R)

4 solutions

(S) continuous for all real x

Final Answer : (D)